

NEOADJUVANT PEPINEMAB ENHANCES IMMUNE CHECKPOINT BLOCKADE IN METASTATIC MELANOMA CHARACTERIZED BY BIOMARKERS OF TME REPROGRAMMING INCLUDING TERTIARY LYMPHOID STRUCTURES

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Background Neoadjuvant immunotherapy is emerging as a critical component in the management of metastatic melanoma. While immune checkpoint inhibitors (ICIs) offer clinical benefit, resistance and disease recurrence remain major limitations, with a substantial proportion of patients failing to achieve pathologic and durable responses. Pepinemab, a novel semaphorin 4D (Sema4D) blocking antibody, reprograms myeloid-driven immune suppression and activates dendritic cells within the tumor microenvironment (TME), offering a complementary mechanism to enhance ICI efficacy.^{1 2}

Methods The phase II integrated biomarker clinical trial (NCT03769155) evaluated neoadjuvant pepinemab in combination with nivolumab and/or ipilimumab in patients with resectable metastatic melanoma. Patients received two preoperative doses, followed by curative-intent surgery and one year of adjuvant nivolumab. Primary study endpoints included multiomic profiling of surgical resected tumor tissue and peripheral blood to assess the immunologic effects of treatment, with a focus on TME modulation and T cell infiltration.

Results Pathologic response was associated with TME remodeling, including neutralization of SEMA4D gradient, increased infiltration of CD4⁺ and CD8⁺ T cells, and the presence of intratumoral B cell aggregates, as demonstrated by multiplex immunohistochemistry (IHC) (figure 1). Bulk tumor

transcriptional profiling revealed enrichment of B cell, T cell, and dendritic cell activation and antigen presentation gene signatures in responders. Peripheral and tumoral biomarkers of DC activation and recruitment were associated with pathologic response and improved recurrence free survival (RFS). Enrichment of non-canonical NF-κB and down-regulation of EMT and MAPK activation pathways were associated with clinical benefit.

Pepinemab-containing regimens induced significantly more mature tertiary lymphoid structures (mTLS) within tumors, defined by multiplex IHC as organized B and T cell zones, CD21⁺ follicular dendritic cells, and CD23⁺ germinal centers (figure 2). mTLS were associated with both pathologic response and RFS. Patients receiving triple therapy exhibited the highest mTLS density, most pronounced immune activation, and greatest clinical benefit, including increased cytotoxic T cell clonal expansion and modulation of cytokine, chemokine, and B cell profiles.

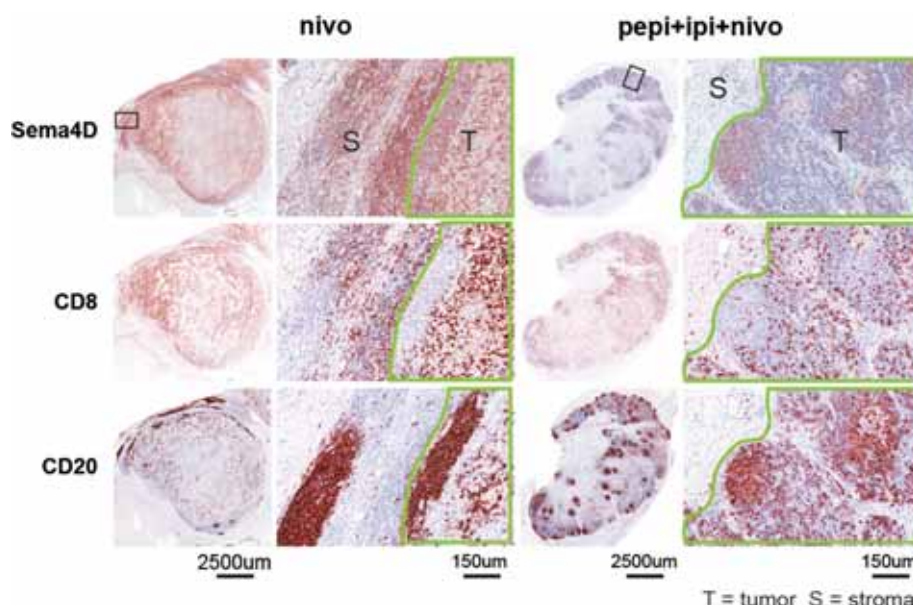
Conclusions Neoadjuvant pepinemab in combination with immune checkpoint blockade promotes robust TME remodeling, overcomes immune suppression, and enhances antitumor immunity through a distinct Sema4D-blocking mechanism. The regimen is safe, well tolerated, and demonstrates promising enhanced clinical activity in resectable metastatic melanoma.

Trial Registration NCT03769155

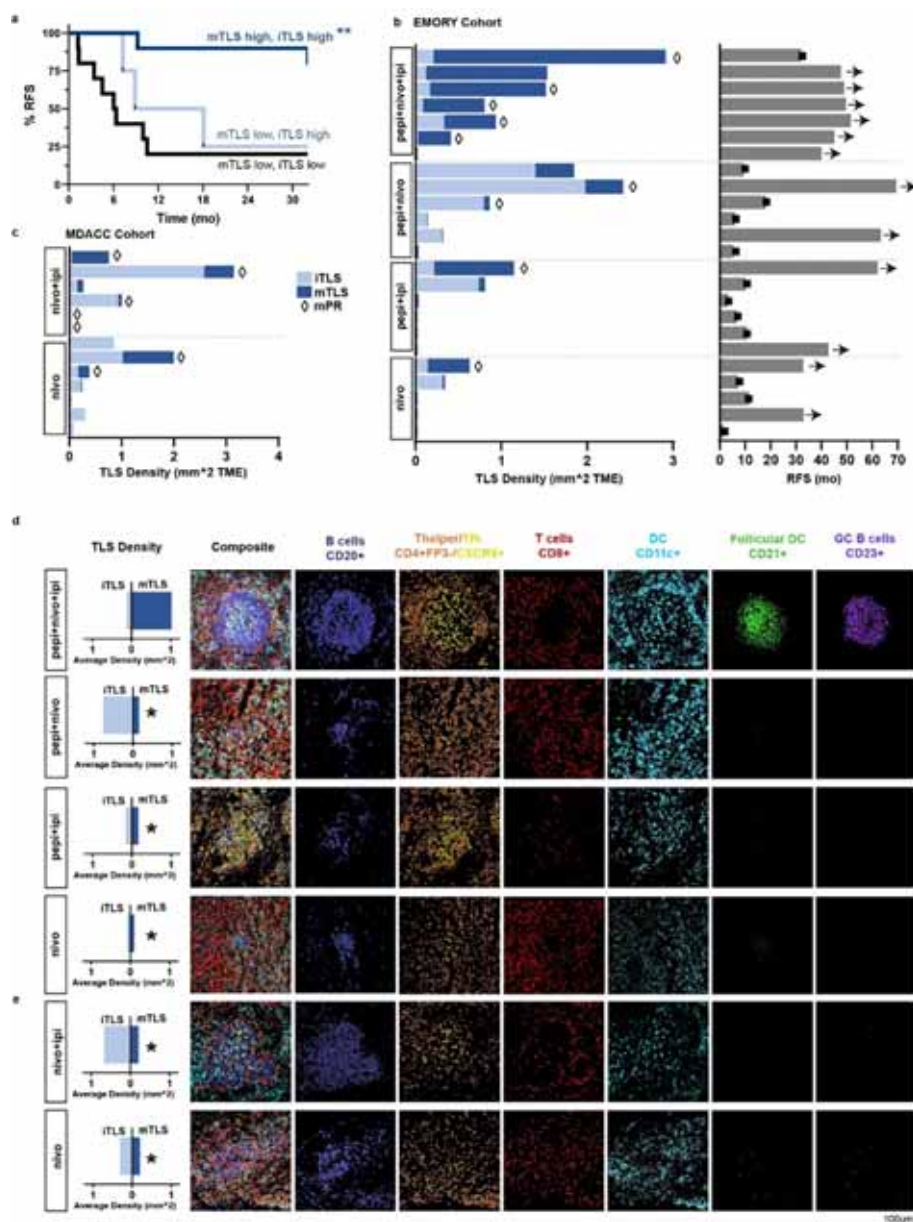
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Ethics Approval This study (Winship 4400-18; PI: Lowe) and its associated secondary use protocol (IRB-00009361; PI: Lesinski) were approved by the Emory University IRB. Written informed consent was obtained for all participants.



Abstract 22 Figure 1 TME remodeling. 1x and 20x magnification of representative images from non-pepinemab and pepinemab-containing post treatment surgical resections



Abstract 22 Figure 2 Pepinemap containing regimens promote formation of mature tertiary lymphoid structures linked to improved clinical outcomes. Immature (iTLS) defined as CD21- CD23- CD20+ aggregates. Mature (mTLS) defined as CD21+/- CD23+ CD20+ aggregates. Squares denote an event (recurrence or death)

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LINEAGE-RESOLVED TRANSCRIPTIONAL PROGRAM DEFINES IMMUNE ENGAGEMENT ARCHETYPES AND PREDICTS IMMUNOTHERAPY RESPONSE ACROSS MULTIPLE CANCER TYPES

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Background Despite the significant clinical benefits of immune checkpoint blockade (ICB), a substantial proportion of patients experience primary or acquired resistance, highlighting a critical need for predictive biomarkers that move beyond descriptive immune readouts. Current approaches often lack the mechanistic depth to fully capture the complex tumor micro-environment (TME) dynamics governing ICB response.

Methods We developed IMIREG (Immune Microenvironment Regulon Signature), a mechanistically grounded 14-regulon signature, by systematically inferring transcription factor (TF) activity across four discovery immunotherapy cohorts (IMvigor010, I-SPY2, Kim, Gide). IMIREG's predictive utility was rigorously validated across 40 independent ICB clinical trial datasets spanning 17 cancer types. We further characterized IMIREG's physiological distribution across healthy tissues and single-cell atlases. Mechanistic dissection involved bulk RNA-seq deconvolution, pathway enrichment analysis, and a novel lineage-resolved classification framework utilizing pan-cancer single-cell RNA-seq compendia to identify immune cell drivers. Spatial immune profiling and longitudinal analyses of a triple-negative breast cancer (TNBC) cohort elucidated IMIREG's dynamics during anti-PD-1 therapy.

Results IMIREG consistently and robustly predicted clinical benefit from ICB (average AUROC ≥ 0.74), outperforming established biomarkers including T cell inflamed signature, IFN- γ signatures, and tumor mutational burden. IMIREG's utility was shown to be predictive, not prognostic. Mechanistically, IMIREG activity was selectively enriched in functionally competent yet checkpoint-restrained immune compartments, notably encompassing exhausted/effector T cells and pro-inflammatory M1 macrophages. IMIREG-high tumors exhibited an inflamed TME with robust antigen presentation and positive correlations with key checkpoint molecules (PD-1, CTLA-4). Our lineage-resolved framework classified IMIREG-high tumors into T cell-driven, macrophage-driven, and dual-driven immune archetypes, revealing substantial inter-tumoral heterogeneity. Strikingly, the most immunologically active dual-driven phenotype exhibited near-complete erosion during metastatic progression. Furthermore, IMIREG served as a dynamic, real-time pharmacodynamic marker in TNBC patients, with higher baseline and on-treatment levels correlating with improved pathological complete response.

Conclusions IMIREG provides a novel, mechanistically grounded biomarker that moves beyond enumerating immune components to measuring the core regulatory circuitry of anti-tumor immunity. This signature significantly enhances patient stratification for ICB and offers a powerful framework to understand the dynamic, multi-lineage nature of immune engagement and its collapse in advanced cancer, enabling more refined immunotherapy strategies.

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LIQUID BIOPSY DETECTS TREATMENT RESISTANCE TO T CELL RECEPTOR (TCR)-T CELL THERAPY

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Background T cell receptor (TCR)-T cell therapy has shown clinical activity in a variety of cancers including melanoma, synovial cell sarcoma, and human papillomavirus (HPV)-associated cancers.¹⁻³ Despite responses in an assortment of malignancies, damaging mutations to human leukocyte antigen (HLA) class I alleles have been identified in non-responding tumors.³⁻⁴ To date, these alterations have been identified through examination of solid tumor biopsies. Practical limitations of tumor biopsy acquisition pose a challenge to studying the frequency and timing of these resistance mechanisms.⁵⁻⁶ Conversely, liquid biopsy offers a minimally invasive and easily repeatable method for real-time monitoring of tumors difficult or inaccessible to biopsy.⁷⁻⁸ This study aimed to determine if HLA-A*02:01 damaging mutations previously identified in tumor biopsies from 2 patients treated in a phase I trial of E7 TCR-T cell therapy for HPV-associated cancers could also be detected with liquid biopsy.³

Methods Archived plasma specimens collected before and after treatment from each of the 2 patients (Patients 4 and 5) were studied. Cell-free (cf)DNA was extracted using the QIAamp Circulating Nucleic Acid kit. Whole exome sequencing (WES) was performed on the Illumina NextSeq200. Somatic damaging mutations were identified by comparing patient-matched germline sequencing to WES of cfDNA using manual curation and the mutation callers mutect, mutect2, vardict and strelka.

Results Patient 4 had primary resistance to E7 TCR-T cell therapy due to a nonsense mutation at amino acid position 51 in the $\alpha 1$ helix of HLA-A*02:01 as determined by a tumor biopsy on day +40 (figure 1, left).³ Liquid biopsy detected this mutation in samples collected at each timepoint: pre-treatment (day -10), early post-treatment (day +10) and progression (day +40). Patient 5 had acquired resistance following a deep partial response with elimination of over 80 tumors. A

progressing tumor specimen from day +349 showed a damaging mutation at amino acid position 141 in the peptide binding region of the $\alpha 2$ helix of HLA-A*02:01 (figure 1, right).³ Liquid biopsy did not detect this mutation pre-treatment (day -12) and at day +150 but did detect it at day +284.

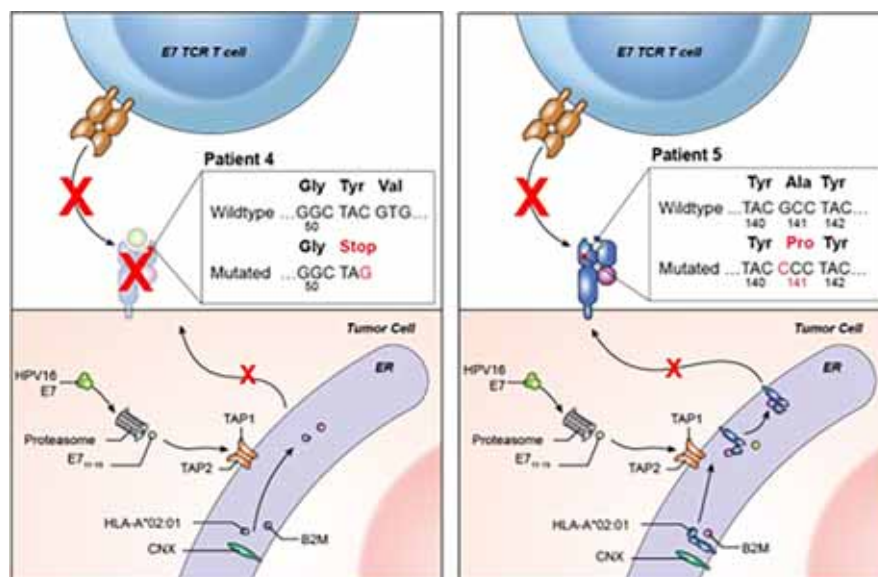
Conclusions These findings highlight the potential for liquid biopsy to detect both primary and acquired resistance to TCR-T cell therapy. This method holds promise for prospectively predicting response and monitoring for the development of resistance to TCR-based therapeutics.

Ethics Approval The trial was registered on clinicaltrials.gov, trial number NCT02858310. This study was approved by the NIH IRB; approval number 16C0154.

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<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0038>



Abstract 38 Figure 1 Detection of damaging mutations in HLA-A*02:01 by liquid biopsy. Illustration depicting somatic damaging mutations in HLA-A*02:01 molecule from treatment resistant tumor for patient 4 (left) and patient 5 (right)

IDENTIFICATION OF THE COMMON INTERPLAY BETWEEN THE IMMUNE SYSTEM AND MULTIPLE CANCERS USING HIGH-THROUGHPUT PLASMA PROTEOMICS

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Background The immune system is posited to play a multifactorial role in cancer, including in the risk for development, growth, and progression of cancer. Moreover, understanding the immune system's interactions with cancer has led to breakthroughs in treatments. However, uncertainties remain in the mechanistic potential of the immune system in the prevention and early detection of cancers, and whether these mechanisms are shared across cancers. To begin to address this gap we assessed the extent of shared immune drivers detectable in the circulation in at-risk individuals across the most common cancer types and explored if potential immune-related pathways exist that may be therapeutically beneficial.

Methods Using the SomaScan Assay™®, approximately 7,000 proteins were assessed in ~15,000 plasma samples in a case-cohort subset of initially cancer-free participants in the European Prospective Investigation into Cancer study. Cox proportional hazards time-to-event univariate models of incident Breast, Lung, Prostate, and Colorectal cancer (n=949, 525, 945, 950 events respectively over 20-year follow-up) identified proteins significantly associated with each incident cancer type using a cutoff of FDR <0.1. Pathway and gene ontology enrichment analyses were conducted using DAVID to determine how the differentially expressed proteins function together in biological systems at the pathway and biological process level, respectively. Enriched pathway/process was considered significant for FDR <0.05.

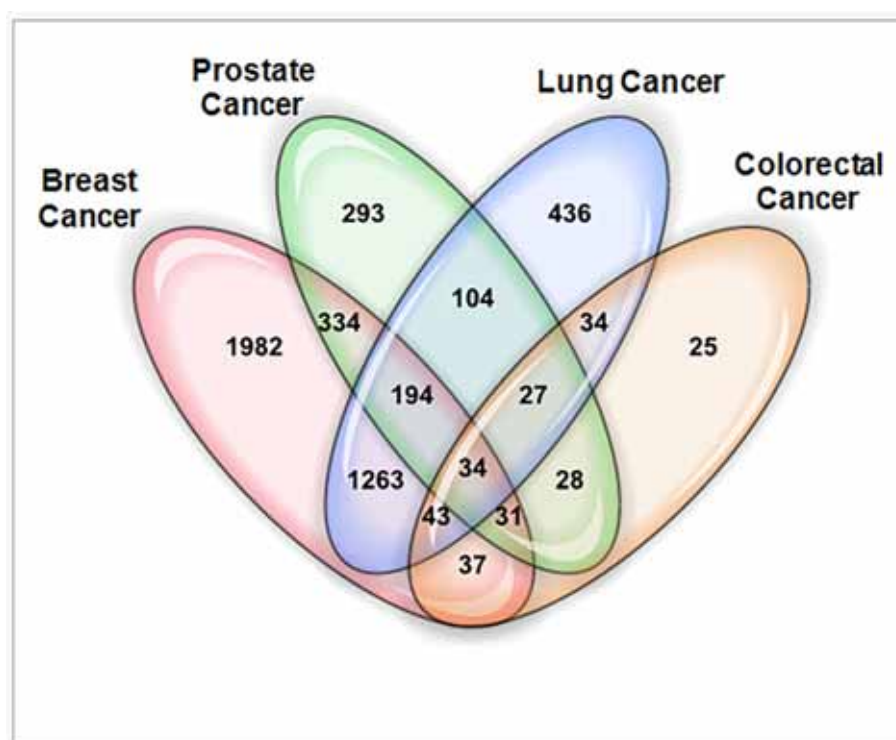
Results 3918, 2135, 1045 and 259 proteins passed the FDR 0.1 cutoff for association with incident Breast, Lung, Prostate and Colorectal cancer, respectively. Lung and breast cancer shared the highest proportion (72%) of overlapping proteins, whereas prostate and lung were the least overlapping cancer types sharing 34% proteins. KEGG pathway analysis of risk-associated proteins identified significant enrichment in immune-related pathways including chemokine signaling (breast and lung), complement and coagulation cascade (colorectal and lung), and Fc gamma R-mediated phagocytosis (lung).

Thirty-four overlapping proteins were identified across all four cancers, with 18 (54%) being involved in at least one immune-related biological process (figure 1). Enrichment analysis indicated that the overlapping proteins converge on Wnt signaling.

Conclusions We identified that blood-based proteomics detects significant immune involvement in the risk for the four most prevalent cancer types. High degrees of proteomic overlap between all cancers that converge on immune-related proteins and functions suggest shared immune interplays across cancers with Wnt signaling pathway as a common druggable or repurposable target. These results grant a better understanding of the common immune drivers across cancers and may aid in providing insights into preventative or early therapeutic targets.

Ethics Approval The EPIC project was approved by the ethical review boards of the International Agency for Research on Cancer (Lyon, France) and from all local centres. Written informed consent was obtained from all participants.

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0039>



Abstract 39 Figure 1 Overlapping risk proteins between common cancer types. The number of unique and common significant proteins (FDR <0.1) associated with future cancer risk

EARLY DYNAMICS IN CIRCULATING TUMOR DNA VERSUS WHOLE BODY FDG-PET/CT PARAMETERS DURING ANTI-PD-1 BASED THERAPY AS A PREDICTIVE BIOMARKER IN ADVANCED STAGE MELANOMA

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Background Evaluation of tumor volume change and [18F]Fluoro-2-deoxy-2-D-glucose (FDG) uptake using whole-body PET/CT are frequently used to evaluate treatment response with immune checkpoint inhibitors (ICIs) in patients with advanced stage melanoma. We previously reported that early circulating tumor DNA (ctDNA) dynamics after ICI initiation appears to be a candidate biomarker to predict treatment response.¹ We hypothesize that early FDG-PET/CT uptake dynamics will offer similar predictive capabilities. Our study explores both early FDG-PET/CT parameters and multi-stages of ctDNA dynamics as predictors of treatment efficacy in ICI-treated patients with advanced melanoma.

Methods A prospective study was conducted for patients with unresectable stage III/IV melanoma who underwent personalized, tumor-informed ctDNA analysis (Natera) and whole-body FDG-PET/CT imaging at baseline and at 3-4 weeks after ICI initiation. Serial ctDNA levels were assessed every 3-4 weeks during treatment. Assessed FDG-PET/CT parameters included: largest target lesion (TaL) volume, TaL SUVmax, TaL SUVmax-to-liver mean ratio, total lesion (ToL) SUVmax, ToL SUVmax-to-liver mean ratio, and ToL SUVmean. Cox proportional hazard models were used to assess associations between each biomarker change from baseline and 6-month progression free survival (PFS) outcome.

Results 16 patients were enrolled and evaluable. Median baseline total number of metastatic lesions per patient was 4. Primary sites of melanoma included: cutaneous (11/16), unknown (2/16), uveal (2/16), mucosal (1/16). Disease stages included: M1d (2/16), M1c (5/16), M1b (4/16), M1a (3/16), III (2/16). ICI regimens included: ipilimumab/nivolumab (8/16), nivolumab/relatlimab (7/16), pembrolizumab (1/16). ctDNA change from baseline to 3-4 weeks was not significantly associated with 6-month PFS (HR 0.998, 95% CI 0.993-1.003, $p=0.408$ for continuous change), but ctDNA change at 6-8

weeks demonstrated significant association with 6-month PFS (HR 0.995, 0.991-0.999, $p=0.022$). Early (3-4 weeks) change of FDG-PET/CT imaging parameters showed stronger predictive signals for PFS. Early decrease in TaL SUVmax-to-liver ratio was strongly associated with 6-month PFS (HR 0.355, 95% CI 0.168-0.753, $p=0.007$ for continuous change; HR 0.038, 0.004-0.361, $p=0.004$ for categorical decrease). Other significant predictors for 6-month PFS included: TaL SUVmax, ToL SUVmax, and ToL SUVmax-to-liver mean ratio (table 1).

Conclusions Early changes in several FDG-PET/CT parameters, particularly SUVmax-to-liver ratio, may be better predictors of PFS than ctDNA dynamics after 3-4 weeks of ICI initiation. Notably, ctDNA dynamics at 6-8 weeks appears to have a strong predictive utility for 6-month PFS. Larger cohorts are needed to optimize the predictability of early ctDNA dynamics and changes in FDG-PET/CT parameters to predict treatment efficacy and long-term progression risk.

Trial Registration NCT06199713

REFERENCE

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Ethics Approval The study protocol (2023-1089) is approved by the University of Wisconsin institutional ethical guidelines and complies with the guidelines of the institutional review board.

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Abstract 90 Table 1 Predictors of 6-month progression free survival (PFS)

Variables (change from baseline)	Continuous change			Categorical decrease		
	HR	95% CI	p-value	HR	95% CI	p-value
ctDNA (Δ 3-4 weeks)	0.998	0.993 – 1.003	0.408	0.269	0.063 – 1.138	0.074
ctDNA (Δ 6-8 weeks)	0.995	0.991 – 0.999	0.022	0.098	0.019 – 0.510	0.006
Target Lesion Volume (Δ 3-4 weeks)	1.001	0.999 – 1.004	0.199	0.053	0.006 – 0.465	0.008
Target Lesion SUVmax (Δ 3-4 weeks)	0.796	0.639 – 0.990	0.041	0.048	0.005 – 0.426	0.006
Target Lesion SUVmax-to-liver mean ratio (Δ 3-4 weeks)	0.355	0.168 – 0.753	0.007	0.038	0.004 – 0.361	0.004
Total Lesion SUVmax (Δ 3-4 weeks)	0.979	0.963 – 0.997	0.019	0.101	0.012 – 0.833	0.033
Total Lesion SUVmax-to-liver mean ratio (Δ 3-4 weeks)	0.959	0.927 – 0.992	0.014	0.111	0.013 – 0.917	0.041
Total Lesion SUV mean (Δ 3-4 weeks)	0.970	0.942 – 0.999	0.040	0.437	0.102 – 1.887	0.265

METABOLIC REPROGRAMMING IN RCC TUMORS CONTRIBUTES TO A DYSFUNCTIONAL IMMUNE RESPONSE WITHIN THE TUMOR MICROENVIRONMENT AND IS CORRELATED WITH DISEASE PROGRESSION

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Background Renal cell carcinoma (RCC), a malignancy arising from renal tubular epithelial cells, represents 2–3% of global cancer diagnoses and 85% of all kidney neoplasms with the most common histological subtype being clear cell renal cell carcinoma (ccRCC), an immunologically and histologically diverse tumor which is associated with a poor clinical outcome. While significant progress has been made in the development of immunotherapy for ccRCC, there are still many unanswered questions about the mechanisms of immune evasion and resistance, and the development of predictive biomarkers for optimal treatment strategies for individual patients. Emerging evidence suggests that metabolic reprogramming characterized by dynamic shifts in nutrient utilization that transcend canonical Warburg physiology to encompass lipid anabolism, nutrient capture and catabolic pathways and microenvironment-driven metabolic plasticity, is key to overall ccRCC pathogenesis. This orchestrated rewiring of cellular dynamics has been suggested to sustain tumor proliferation under hypoxia while fostering immunosuppression through metabolite-mediated T cell exhaustion and potentially the presence of tumor associated macrophages (TAMs).

Methods To investigate the complex interplay between the immunogenic nature of ccRCC tumors and metabolic reprogramming, we performed both gene expression and spatial proteomic profiling of 16 ccRCC patient samples representative across all clinical TNM stages (Stages I-IV) to define the changes in metabolic processes and infiltrating immune cells correlated with advancing disease. Specifically, we incorporated the *new* PaletteTM end to end spatial workflow (NeoGenomics Laboratories, Inc.) and developed a panel that includes: CD3, CD4, CD8, FoxP3, CD68, CD80, CD163, CD206, CA9, CD31, LAG3, PD1, PanCK. We also included analysis using the analytically validated NanoString nCounter[®] Metabolic Pathways Panel to investigate metabolic reprogramming, cellular stress, and the relationship to the composition of the TME in advancing disease progression in ccRCC.

Results and Conclusions We identified significant changes associated with the TME, especially immunosuppressive immune cells in advanced ccRCC disease, coupled to the identification of several metabolism-related genes that correlated with immune infiltrates in ccRCC. Identifying and understanding these metabolic alterations within the different staging of RCC will help in designing new targeted therapies and improving diagnostic tools to better care for patients with this disease.

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COMBINATION OF ADOPTIVE CELL TRANSFER WITH β/γ -SPECIFIC INTERLEUKIN-2 THERAPY

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Background Adoptive cell therapy (ACT) using ex vivo-expanded autologous tumor-infiltrating lymphocytes (TILs) has shown the potential to induce long-lasting remissions in melanoma. Recent clinical trials have validated the efficacy of TIL-ACT, leading its regulatory approval for patients with metastatic melanoma after failure of first-line immune checkpoint inhibitors. While high-dose interleukin-2 (IL-2) enhances efficacy, it is associated with considerable toxicity such as fever/chills, hypotension and pulmonary complications. Moreover, IL-2 may expand regulatory T cells (Tregs) through its high-affinity α -receptor, potentially limiting efficacy. ANV419, an IL-2 variant selective for the IL-2 receptor β/γ chains, preferentially targets effector T and NK cells, offering extended half-life and an improved safety profile.

Methods The BaseTIL-03M trial (NCT05869539) is a phase I study evaluating the safety and preliminary efficacy of combining TIL-ACT with ANV419 in patients with metastatic cutaneous melanoma refractory to standard systemic therapies. Ten patients received TILs in combination with ANV419 (243 mcg/kg) administered on the day of TIL transfer and again 14 days later. Prior to TIL infusion, patients underwent lymphodepletion with cyclophosphamide and fludarabine. The primary endpoint was safety within the first 30 days post-TIL infusion. Secondary endpoints included overall response rate (ORR), progression-free survival (PFS) and overall survival (OS). In parallel, extensive translational studies of T cells blood and tumor tissues are underway.

Results All ten patients were enrolled between June 2023 and May 2025. No dose-limiting toxicities were observed. Full safety data, ORR, and interim analyses of PFS and OS for the cohort will be presented at the meeting. Additionally, findings from translational studies including T cell repertoire profiling, single-cell proteomics of the TIL products and peripheral blood, and multiplex tissue imaging before and after treatment will be shared.

Conclusions The BaseTIL-03M trial demonstrates that combining TIL-ACT with ANV419 is feasible and well tolerated, with no dose-limiting toxicities observed and preliminary signals of clinical benefit. These encouraging findings support advancement to a phase II trial focused on efficacy, to be conducted with the PragmaTIL network.

Trial Registration Clinical TrialsGov: NCT05869539

Ethics Approval All patients were consented and signed the ICF before being included in the trial.

Consent Written informed consent was obtained from the patient for publication of this abstract and any accompanying images. A copy of the written consent is available for review by the Editor of this journal.

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KLF2-OVEREXPRESSION IMPARTS STEMNESS ONTO DYSFUNCTIONAL NEOANTIGEN-REACTIVE HUMAN TUMOR-INFILTRATING LYMPHOCYTES

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Background Adoptive cell transfer (ACT) of tumor-infiltrating lymphocytes (TILs) targeting patient-specific neoantigens can mediate durable tumor regressions in metastatic epithelial cancers. In melanoma, frequency of neoantigen-reactive stem-like TIL (TIL_{SL}) (CD39-CD69-) within infusion products correlate with better clinical outcomes. Antitumor TIL_{SL} could serve as an ideal cell therapy, since they self-renew, differentiate into effectors, lack inhibitory proteins of exhaustion, and persist in patients post-ACT. However, in most solid tumors, only a minority of antitumor neoantigen-TIL exist in TIL_{SL} state, while the vast majority are CD39+ terminally-differentiated, dysfunctional, exhausted (TIL_{EX}) states with poor proliferative, and antitumor effect. Therefore, a goal for enhancing TIL-ACT is to identify, or engineer stem-like antitumor T cells to enhance cell therapy response.

Methods The transcription factor Krüppel-like factor 2 (*KLF2*) is enriched in ACT response-associated stem-like neoantigen TILs in melanoma. We investigated the impact of *KLF2*-overexpression on patient-derived, neoantigen-reactive CD8+ TILs from multiple epithelial cancers. Flow cytometry, single-cell transcriptomics, and T cell receptor (TCR) clonotype tracking were used to analyze changes in T cell phenotype and clone frequency. *In vivo* antitumor efficacy of *KLF2*-engineered T cells was evaluated using human TP53-R175H TCR-transduced T cells in an ovarian cancer cell-line model in immunodeficient mice.

Results *KLF2* overexpression in dysfunctional neoantigen-reactive CD8+ TIL_{EX} showed a significant increase of the stem-like TIL_{SL} population (CD39-/CD69-, 3-80-fold, *P*=0.001, paired T-test) and decrease of the exhausted population (CD39+/CD69+, 1-16-fold, *P*=0.017, paired T-test). Single cell transcriptomic and TCR analysis of antitumor or neoantigen-reactive TIL from 6 patients demonstrated that *KLF2*-engineered selected TIL exhibited high enrichment of TIL_{SL} transcriptional signature and consistent reduction in TIL_{EX} transcriptional signature, even in monoclonal antitumor TIL. Single cell TCR-clonotype tracking studies of 40 selected TIL clones in two patients demonstrated that 11 out of 40 (27.5%) clones enriched exclusively in CD39+ states were detected in CD39- stem-like state after *KLF2*-expression. Lastly, human T cells transduced with *KLF2* and TP53-R175H TCR exhibited significant tumor regression of human TYK-nu ovarian cancer *in vivo*, even in the very low cell dose of 6e5 cells/mouse, when compared to TP53-R175H TCR-transduced cells (*P*=0.025, 2way ANOVA, *n*=5 mice per group).

Conclusions *KLF2*-overexpression in neoantigen-reactive human T cells results in a phenotypically distinct population, enriched with stem-like signature and exhibiting enhanced antitumor activity. These results could potentially improve *KLF2*-engineered neoantigen-reactive stem-like T cells for ACT targeting human cancer.

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SELECTION OF CD103+CD31-CD8+ CELLS ENRICHES TUMOR REACTIVITY OF TUMOR INFILTRATING LYMPHOCYTES (TIL) INDEPENDENT OF PERSONALIZED SEQUENCING

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Background Adoptive transfer of autologous T cells from tumor fragments cultured in IL-2 has been successful in treating melanomas and select epithelial cancers.^{1 2} However, tumor-reactive T cells are a small percentage of all TIL while most are bystanders.³ Identification of neoantigen-reactive TIL for treatment involves time and resource-intensive testing of TIL cultures against patient-specific candidate neoantigens, resulting in selection of TIL at the population rather than cell level.⁴ Use of cell surface markers to isolate tumor-relevant TIL will circumvent the need for personalized sequencing and screening and generate a more polyclonal TIL product.

Methods In this study, we cultured multiple fragments from resected tumors, together in GREX100 flasks with high dose IL-2 (n=9) obtaining a median of 4.4e8TIL. The large cell yield allowed the study of cell surface marker expression for sorting tumor-reactive TIL by FACS. To identify candidate markers, we performed single-cell RNA-, CITE- and TCR sequencing on TIL from a human rectal cancer lung metastasis. We cloned and expressed high frequency CD8TCRs in donor lymphocytes and screened them for tumor reactivity. Twelve TCRs showed functional reactivity to tumor neoantigens and 3 were reactive to tumor-irrelevant CMV/EBV/Flu peptides (indicating true bystanders). Projecting relevant TCRs onto the transcriptomic map indicated that cells with tumor-reactive TCRs clustered separately from bystanders. Antibody expression analysis showed CD103 as the top marker for reactive cells and CD31 for bystanders. Subsequently we evaluated these markers to isolate tumor-reactive CD8+TIL.

Results Among the 9 patients studied, the percentage of CD103+CD31- cells ranged from 4.6%-44% among CD8 +TIL. TIL sorted based on CD103/CD31 expression were expanded using the rapid expansion protocol. Subsequent screens against candidate neoantigens demonstrated enriched reactivity among CD103+CD31-CD8+TIL in 7/9 patients compared to bulk CD8+T cells. Mean functional reactivity (upregulation of 4-1BB following on-target stimulus) among the CD103+CD31-CD8+TIL was 20.7% and median fold enrichment was 5.7X over bulk CD8+TIL. Single-cell TCR sequencing analysis performed on the reactive T cells enabled identification and validation of 1-3 unique neoantigens per patient, in some cases with multiple clonotypes against the same neoantigen. Further, sorting CD103+CD31-CD8+T cells enriched for cells with stem-like phenotype⁵ indicating their enhanced anti-tumor potential.

Conclusions This study has strong implications for the identification of reactive CD8+TIL from bulk cultures of tumor fragments without requiring patient-specific testing. Pending successful validation of scalability of this approach under GMP conditions, it could substantially decrease time-to-treatment (~5 weeks) making continued exploration of TIL therapy viable for more epithelial cancer patients.

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Ethics Approval This study was approved by National Institutes of Health (NIH) Internal Review Board (IRB); approval number NCT00068003.

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CUTANEOUS ERUPTIONS ASSOCIATED WITH LIFILEUCEL, A FIRST-IN-CLASS TUMOR-INFILTRATING LYMPHOCYTE THERAPY, AND INTERLEUKIN-2 IN INDIVIDUALS WITH METASTATIC MELANOMA: A RETROSPECTIVE PROGNOSTIC ANALYSIS

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Background In the C-144-01 phase II trial of lifileucel – a first-in-class autologous tumor-infiltrating lymphocyte (TIL) therapy recently approved for advanced/metastatic melanoma – and up to six interleukin-2 (IL-2) doses,^{1,2} ‘rash’ was a treatment-emergent adverse event in 37.2% of individuals.³ TIL/IL-2-associated eruptions have not yet been described or associated with tumor response.

Methods The MGB/DFCI Research Patient Data Registry was queried for patients who received ‘lifileucel,’ ‘Amtagvi,’ or ‘LN-144.’ The lifileucel admission and oncology follow-up encounters were reviewed for demographics, eruption features, dermatopathology, and oncologist assessments of tumor response on 30-day post-TIL restaging scans.⁴ A pre-specified crude sensitivity analysis tested for independent association between IL-2 dosing (‘high IL-2’ as 4-6 doses; ‘low IL-2’ as 1-3 doses), rash, and 30-day response. Binomial family linear regressions modeled tumor response as a binary outcome, with development of lifileucel/IL-2 skin toxicity as a binary predictor. Adjustments were made for age, sex, IL-2 dose number, and TIL delivery-to-discharge days.

Results Among 44 individuals (34.1% female; median age 57 years), 22 (50.0%) developed a lifileucel-associated cutaneous eruption (table 1). Eruptions were asymptomatic or mildly pruritic central-predominant morbilliform eruptions with occasional purpuric features (figure 1A-F). Dermatopathology from

a representative case demonstrated a lymphocyte-predominant interface infiltrate with rare eosinophils and dermal mucin (figure 1G-J).

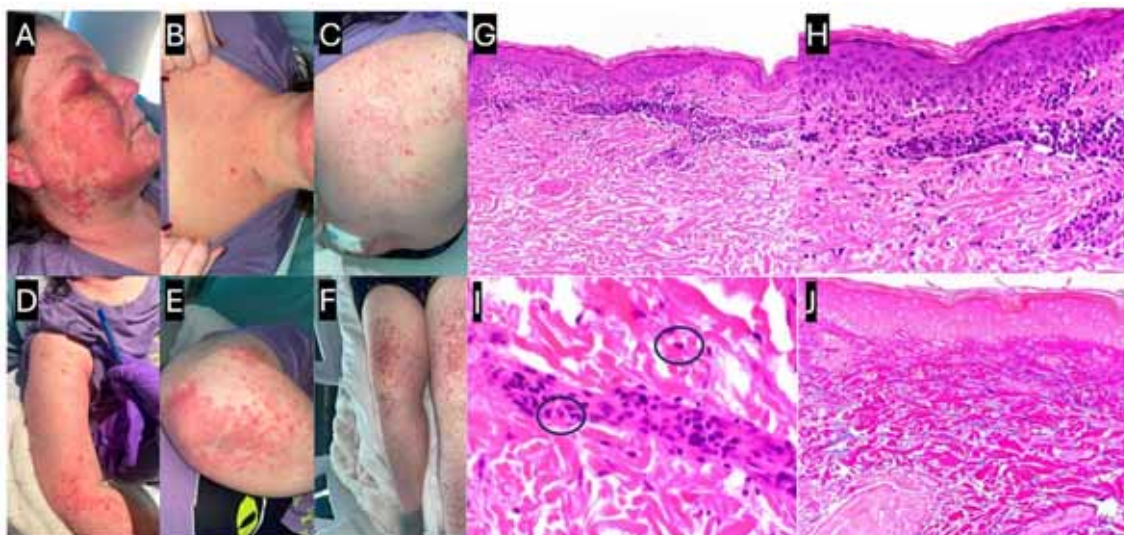
Individuals with cutaneous eruptions had a significantly higher 30-day ORR than those who did not develop cutaneous eruptions (68.2% vs 27.3%, $p=0.01$). Female sex also was descriptively associated with an increased ORR (80.0% vs 31.0%, Fisher’s $p=0.004$). Individuals who developed skin toxicity received more mean IL-2 doses (4.77 vs 3.64, $p=0.02$); however, ‘high IL-2’ recipients did not more frequently demonstrate response (high IL-2: 50.0% vs low IL-2: 43.8%, $p=0.76$).

In the unadjusted logistic regression, lifileucel/IL-2 skin toxicity was associated with tumor response (OR 5.71, 95%CI 1.63–22.47, $p=0.009$). This association was durable when adjusted for IL-2 dose number (aOR 7.21, 95%CI 1.81–36.28, $p=0.008$), and when adjusted for age, sex, TIL delivery-to-discharge days, and IL-2 dose number (a’OR 11.59, 95%CI 2.06–100.59, $p=0.011$).

Conclusions This study demonstrates a positive prognostic association between lifileucel-associated morbilliform eruptions and 30-day ORR. While IL-2 monotherapy can cause similar morbilliform eruptions,^{5,6} this IL-2-dose-adjusted analysis suggests that lifileucel-associated skin toxicity may represent a non-IL-2-mediated marker of underlying anti-tumor efficacy, assessable during the lifileucel-associated hospital admission 3–4 weeks prior to 30-day restaging. Limitations include sample size, retrospective data, and selection bias favoring more pronounced cutaneous eruptions; we recommend future analyses investigating further-out response endpoints.

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Abstract 380 Figure 1 Clinical photography and dermatopathology of lifileucel/interleukin-2-associated morbilliform eruption. **A)** Face. **B)** Chest. **C)** Back. **D, E)** Bilateral arms; left arm biopsy site. **F)** Bilateral legs. **G-H)** Interface dermatitis (H&E stain, 200x and 400x). **I)** Rare eosinophils (circled) (H&E stain, 600x). **J)** Colloidal iron stain with dermal mucin (400x)

BIOMARKER INSIGHTS AND LONG-TERM OUTCOMES OF TILT-123 ONCOLYTIC ADENOVIRUS COMBINED WITH TUMOR-INFILTRATING LYMPHOCYTES IN REFRACTORY METASTATIC MELANOMA: RESULTS FROM THE TUNINTIL PHASE 1 TRIAL

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Background Tumor-infiltrating lymphocyte (TIL) therapy was recently approved by the FDA for metastatic melanoma (MM), but its widespread implementation remains limited by significant toxicities associated with lymphodepleting chemotherapy and high-dose IL-2 regimens. Phase 1 trial TUNINTIL (NCT04217473) evaluated a novel design where TILT-123 (igrelimogene litadenorepvec), an oncolytic adenovirus engineered to selectively replicate in cancer cells and produce tumor necrosis factor (TNF) and interleukin-2 (IL-2), replaces these toxic pre- and post-conditioning regimens.

Methods Seventeen heavily pretreated (range of 1 to 7 prior cancer treatment lines) patients with immune checkpoint inhibitor (ICI)-refractory MM (8 cutaneous, 5 mucosal, 4 uveal) received TILT-123 intravenously and intratumorally in a dose-escalating manner across five cohorts. TILs were administered without preconditioning chemotherapy or post-infusion IL-2. Patients were evaluated for safety (primary endpoint), treatment response and survival outcomes. TILs phenotyping, neutralizing antibody titers in serum, immunohistochemistry of tumor biopsies, serum cytokine measurements, and viral shedding evaluation were performed to better understand the biological mechanisms of response.

Results Treatment was well-tolerated with no dose-limiting toxicities. Most adverse events were mild to moderate, with fever (65%), fatigue (29%), and nausea (29%) being most common. PET imaging revealed 47% (7/15) disease control and 27% (4/15) complete and partial metabolic responses overall, and 40% responses on highest dose level. With RECIST1.1, the objective response rate was 11.7% (2/17) with disease control seen in 35% (6/17). Median overall survival was 447 days with 6 patients surviving more than 600 days. One mucosal melanoma patient is in complete response for more than 4 years.

Viral DNA was detected in both injected and non-injected tumors, with increased T cell infiltration observed at both sites, suggesting effective systemic response. Patients with detectable virus in tumors demonstrated prolonged survival ($p=0.0441$). Interestingly, higher neutralizing antibody titers early in treatment correlated with improved disease control ($p=0.0062$). Moreover, decreased blood lymphocyte counts following TILT-123 administration (i.e. trafficking to tumors) was associated with better tumor control.

Serum proteomic analysis revealed that patients with disease control exhibited a distinct inflammatory signature after initial

viral treatment, suggesting that differential immune responses to TILT-123 may influence subsequent TIL therapy efficacy.

Conclusions TILT-123 combined with TILs demonstrates good safety and promising efficacy in ICI-refractory melanoma, successfully eliminating toxic conditioning regimens while achieving durable responses, including in challenging non-cutaneous subtypes. This novel combination approach deserves continued clinical development as a potentially safer alternative to conventional TIL therapy. A phase 1b trial is ongoing (NCT06961786).

Trial Registration NCT04217473

Ethics Approval The trial was approved by the Danish National Ethics Committee and the Danish Medicines Agency with the approval number 1905760. All approvals were obtained prior to inclusion of the first patient in the trial.

Consent Written informed consent was obtained from the patient for publication of this abstract and any accompanying images. A copy of the written consent is available for review by the Editor of this journal.

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TUMOR AGNOSTIC SPATIALLY DEFINED IMMUNE CELL GENE SIGNATURES PREDICT IMMUNOTHERAPY OUTCOMES

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Background Immune checkpoint inhibitors (ICIs) have shown clinical efficacy across multiple tumor types, including melanoma, non-small cell lung cancer (NSCLC), head and neck squamous cell carcinoma (HNSCC), and triple-negative breast cancer (TNBC), particularly in tumors with high PD-L1 expression.^{1–4} While tumor-intrinsic characteristics differ across cancer types, immune programs may exhibit strong cross-cancer similarity, indicative of shared immune-modulatory mechanisms. We hypothesized that conserved features of the tumor microenvironment, especially from immune cell populations such as CD45+ leukocytes and CD68+ macrophages, could serve as tumor agnostic predictors of ICI response across solid tumors.

Methods We collected spatial transcriptomic data from seven cohorts spanning four cancer types (n = 314): melanoma; n=100, NSCLC; n=117, TNBC; n=42, and HNSCC; n=55. Each cancer type included two independent ICI-treated cohorts, except for HNSCC. We extracted spatially resolved gene expression data from CD45+, CD68+, and CK/S100B+ (tumor) compartments. K-means clustering was used to assess cross-cancer similarity of cell-type-specific profiles. Pairwise cosine similarity scores were computed from gene expression profiles per compartment, and within- versus between-cancer similarities were compared using the Wilcoxon rank-sum test. Predictive gene expression signatures were trained on one cohort per cancer type and half of the HNSCC cohort using a LASSO-based predictor trained on differentially expressed genes (50% train/test split).

Results Clustering of gene expression profiles revealed clear separation between tumor and immune compartments, with cell types forming distinct, compartment-specific groupings across cancer types (figure 1A). K-means clustering revealed that clusters were primarily defined by cell type and, to a

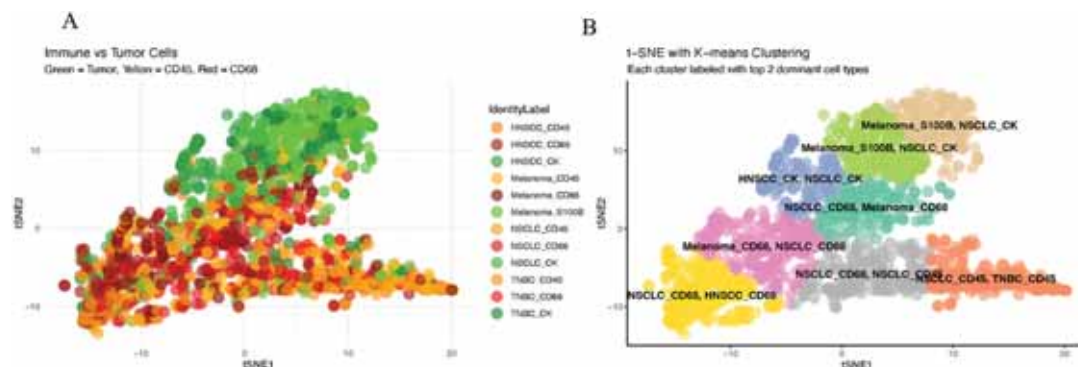
lesser extent, by cancer type (Figure 1B). Tumor and CD45 compartments showed significantly higher gene expression similarity within the same cancer type compared to across cancer types (tumor: $p < 2.2 \times 10^{-16}$, CD45: $p = 4.8 \times 10^{-4}$), whereas CD68 did not show significance ($p = 0.31$), suggesting that CD68 gene expression profiles are more conserved across cancer types. Gene expression signatures from each compartment predict immunotherapy outcomes across cancer types achieving AUCs of 0.95 (95% CI: 0.93–0.97) for CD68, 0.64 (95% CI: 0.56–0.71) for CD45, and 0.92 (95% CI: 0.89–0.95) for tumor in training; and 0.81 (95% CI: 0.74–0.86), 0.63 (95% CI: 0.55–0.72), and 0.65 (95% CI: 0.57–0.92) respectively in testing ($p < 0.01$ for all) (figure 2).

Conclusions These results support our hypothesis that compartment-specific gene expression, particularly CD68 subset, may allow generation of a spatially informed tumor agnostic gene signature that can predict immunotherapy outcomes across diverse cancer types.

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Ethics Approval Resected tissue samples were collected retrospectively with approval from the Yale Human Investigation Committee protocol #9505008219. Written informed consent, or waiver of consent in some cases, was obtained from Yale cohort patients with the approval of the Yale Human Investigation Committee. For the neoadjuvant clinical trial (NCT02489448), Ethical approval was obtained from the Yale Human Investigations Committee (HIC; Yale University, New Haven, CT; HIC no. 1409014537)



Abstract 515 Figure 1 Clustering of spatial transcriptomic profiles across three compartments in four cancer types. (A) t-SNE plot of gene expression, colored by compartment: Tumor (green gradient), CD45+ (yellow gradient), CD68+ (red gradient). (B) t-SNE with k-means clustering of all ROIs (2/patient), annotated by top two cell types per cluster

PRIMARY ANALYSIS OF PATIENTS WITH HIGH-RISK LOCALIZED, LOCALLY RECURRENT, OR REGIONALLY ADVANCED CUTANEOUS SQUAMOUS CELL CARCINOMA (CSCC) TREATED WITH NEOADJUVANT PD-1 THERAPY (NEOPOWER)

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Background PD-1 blockade achieves response rates of ~50% in unresectable/metastatic cutaneous squamous cell carcinoma (CSCC) with durable responses lasting >6 months in 60% of patients. Here we present primary results from a phase 2, multicenter, open-label study of neoadjuvant cemiplimab in patients with high-risk, resectable CSCC (NCT04315701).

Methods The NeoPOWER study enrolled patients with resectable CSCC, according to 1 of the following categories: 1) high-risk localized CSCC (tumor diameter >2.0 cm, depth>6mm, poorly differentiated histology, or perineural invasion), 2) recurrent CSCC after prior surgery/radiation, or 3) regionally advanced CSCC (in-transit, subcutaneous or nodal metastases). Patients received cemiplimab 350mg IV every 3 weeks for 3 cycles, then underwent definitive surgery; no adjuvant therapy was administered. The primary endpoint was pathologic partial response (PPR); secondary endpoints included pathologic complete response (PCR), objective response rate (ORR), 12 month progression free survival (PFS) and toxicity.

Results Among 35 patients enrolled, the median age was 72 yrs (range 41-95), 74% were male, 63% were white; all patients (100%) were ECOG 0 or 1. There were 18 (51%) patients with high risk localized CSCC, 3 (9%) with recurrent CSCC, and 12 (34%) with regionally advanced CSCC; 40% of patients had nodal involvement. Most patients (91%) received all 3 cycles of cemiplimab; 28 patients (80%) completed definitive surgery. The median duration of follow-up was 9.9 months (0.7-28.5). At the time of reporting, 28 patients (82%) had completed study treatment; 2 (6%) stopped due to progression, 2 (6%) discontinued due to toxicity, and 1 patient withdrew consent after 1 cycle of therapy. Among 29 patients with available pathology, the overall PPR rate was 89.7%, including PCR for 72.4% and near PCR (NPCR) 10.3%. The RECIST ORR was 58.8%, with 50.0% partial response (PR) and 8.8% complete response (CR); 29.4% had stable disease (SD). The 12 month PFS rate was 82%; median progression free and overall survival were not reached (NR). TRAE included low-grade fatigue (22.9%), increased AST (11.4%), anorexia (8.6%), pruritus (8.6%), and maculopapular rash (8.6%). There were no grade 3-5 TRAE reported.

Conclusions Neoadjuvant cemiplimab for 3 cycles prior to surgery is well tolerated and highly effective with PPR 89.0% and PCR 72.4% in resectable CSCC, despite a lower clinical/radiographic RECIST ORR of 58.8%. Neoadjuvant PD-1 therapy should be considered a potential new treatment paradigm for patients with high risk localized, recurrent or regionally advanced, resectable CSCC.

Acknowledgements This investigator-initiated clinical trial was supported by Regeneron Pharmaceuticals. The authors gratefully acknowledge Regeneron and their clinical trials research team for their support, including the provision of study funding and investigational drug supply.

Ethics Approval This clinical trial was approved by the Institutional Review Board (IRB) at the University of Southern California. The study was conducted in accordance with the ethical standards of the IRB and the principles outlined in the Declaration of Helsinki. The official IRB approval number for this study is HS-19-00848. Written informed consent was obtained from all individual participants prior to their enrollment in the trial.

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REAL-WORLD EFFECTIVENESS OF NEOADJUVANT IMMUNOTHERAPY IN RESECTABLE STAGE III-IV MELANOMA: A COMPARATIVE ANALYSIS OF NIVOLUMAB/IPILIMUMAB VERSUS PEMBROLIZUMAB

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Background The use of neoadjuvant/perioperative immunotherapy in the NADINA and SWOG S1801 trials has shown improved treatment outcomes in stage III cutaneous melanoma compared to adjuvant therapy. However, real-world data (RWD) on neoadjuvant immunotherapy remain limited.

Methods We retrospectively analyzed 52 patients with resectable stage III-IV cutaneous melanoma treated with neoadjuvant immunotherapy at a single academic center. Patients received either nivolumab + ipilimumab (n=21) or pembrolizumab monotherapy (n=31). Radiological responses were assessed using RECIST v1.1, and pathological responses were classified as major pathological response (MPR, $\leq 10\%$ viable tumor) and complete pathological response (pCR). Patients receiving combination therapy who achieved a major pathological response (MPR) did not undergo further adjuvant treatment, whereas those treated with pembrolizumab continued it in the adjuvant setting for a total treatment duration of up to one year. Survival outcomes were estimated using Kaplan-Meier methodology.

Results Radiological objective response rates (ORR) were similar in both groups: 57.1% for nivolumab/ipilimumab and 51.6% for pembrolizumab ($p=0.78$). However, pathological responses favored combination therapy: MPR was achieved in 71.4% vs. 32.2% ($p=0.01$, table 1); pCR in 42.6% vs. 29.0% (table 1). Grade 3 immune-related toxicity occurred in 19% (combination therapy) vs. 12.9% (monotherapy). Radiological-pathological concordance (CR/PR vs. MPR) was 68.9% ($\kappa = 0.368$).

Trend-level associations were observed for baseline body weight (OR 1.35 per 10 kg, $p=0.065$) and treatment regimen (nivolumab/ipilimumab OR 36.78, $p=0.061$) with MPR.

Median progression-free survival (PFS) was not reached after 16 months of follow-up. One- and two-year PFS rates were higher in the combination group (87% at both time-points) versus pembrolizumab (77% and 68%, respectively). Achieving MPR was associated with significantly better PFS ($p=0.026$): 94% 2-year PFS with MPR vs. 61% without.

Conclusions The use of combination immunotherapy regimens is associated with higher rates of pathological response and a trend toward improved progression-free survival (PFS) compared to monotherapy. Moreover, achieving a major pathological response (MPR) may allow for de-escalation of systemic

therapy after surgery, despite the increased treatment-related toxicity observed with combination approaches.

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Abstract 525 Table 1 Pathological response

	Nivolumab Ipilimumab(n=21)	Pembrolizumab (n=28/31)
MPR (○)	15 (71.4%)	10 (32.2%)
pCR (●)	9 (42.6%)	9 (29.0%)
near pCR (●)	6 (28.6%)	1 (3.6%)
pPR (○)	1 (4.8%)	4 (14.3%)
pNR (●)	4 (19.0%)	14 (50.0%)

PRIMARY RESULTS WITH LONG-TERM FOLLOW UP OF A PHASE II STUDY OF NEOADJUVANT AND ADJUVANT CEMIPIMAB IN HIGH-RISK RESECTABLE CUTANEOUS SQUAMOUS CELL CARCINOMA

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Background Cemiplimab was approved for treatment of metastatic cutaneous squamous cell carcinoma (CSCC) based on the EMPOWER-CSCC-1 trial. A phase II study of 79 patients receiving four doses of neoadjuvant cemiplimab for resectable CSCC showed a major pathologic response (MPR) rate of 64%. In a synchronous study we hypothesized that peri-operative cemiplimab would lead to high response rates and excellent recurrence-free survival (RFS).

Methods Winship4851 is a single-arm phase II study of neoadjuvant and adjuvant cemiplimab for high-risk resectable CSCC defined as: nodal disease with extracapsular extension or one node ≥ 20 mm; in transit metastases; T4 head and neck (H&N) tumor; or recurrent CSCC (NCT04428671). Patients received three doses of cemiplimab every three weeks followed by surgery and 18 adjuvant doses of cemiplimab until recurrence or toxicity. Adjuvant radiation was offered at the discretion of the investigator. Primary endpoint was pathologic response rate. Secondary endpoints were safety and feasibility, RFS and overall survival (OS). The study was approved by Emory University's IRB.

Results Fourteen patients were enrolled; the median age was 73 (range 53-87) and 13 (92.8%) were males. Thirteen (92.8%) had H&N disease; one had axillary nodal recurrence. All patients completed neoadjuvant therapy except one who developed grade 2 arthralgias, and all patients underwent

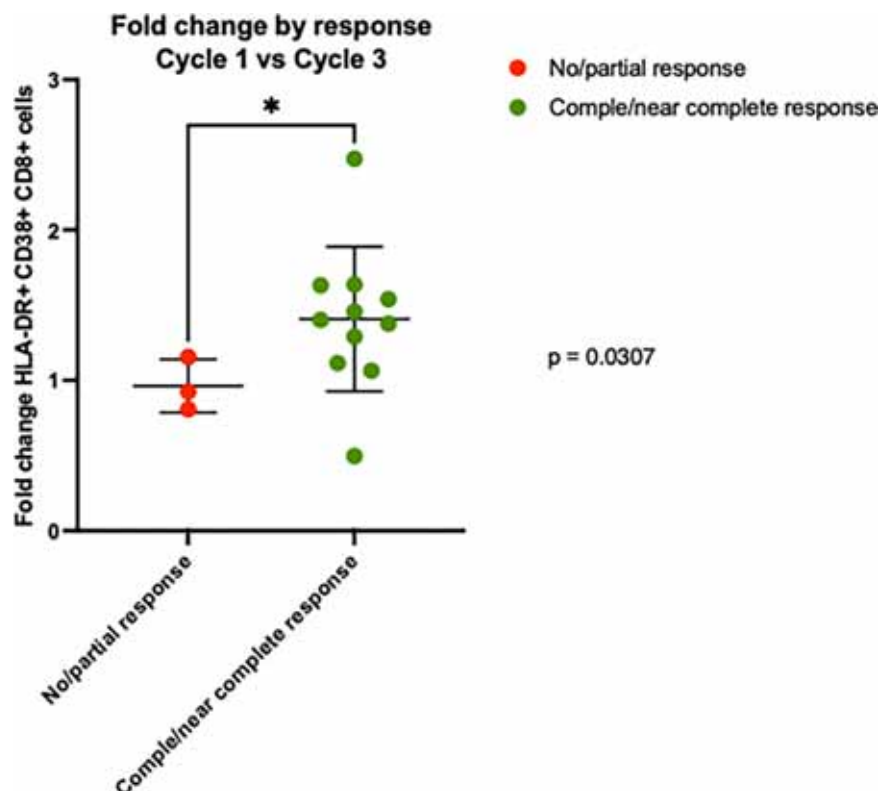
planned curative-intent resection. Rates of complete, near-complete, and partial pathologic responses were 57.1%, 21.4%, and 14.3%, respectively, with major pathologic response (MPR) rate of 78.5%. Four patients underwent adjuvant radiation, two with MPR and two with non-MPR. All four of these patients were treated by the H&N team; the remaining 10 patients treated by the cutaneous malignancy team did not undergo radiation. Median time to adjuvant cemiplimab was 32 days (range 15-50) in patients that did not undergo radiation. At median follow up of 33.4 months, one patient with pathologic non-response experienced recurrence and died; one patient died without disease. Three-year RFS and OS rates were 92.3% and 85.7%. One patient experienced grade 3 pneumonitis that developed after adjuvant treatment was complete. Patients with MRP had significantly higher increase in HLA-DR+ CD38+ CD8+ T cells from baseline compared to pathologic non-responders (figure 1).

Conclusions Peri-operative cemiplimab is safe and well-tolerated and leads to excellent pathologic responses with only three neoadjuvant doses of cemiplimab. Use of adjuvant radiation was disease-team dependent and in this small study did not appear to correlate with recurrence. Further studies assessing the impact of adjuvant radiation and/or systemic therapy are warranted.

Trial Registration NCT04428671

Ethics Approval The study was approved by the Emory IRB (approval number IRB00115160) and participants gave informed consent to take part in this study.

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0527>



Abstract 527 Figure 1 Fold change by response: cycle 1 vs. cycle 3

A NOVEL MULTIPLEX IMAGING-BASED IMMUNOTHERAPY PANEL AND AI-POWERED ANALYSIS SOLUTION IDENTIFIED PREDICTIVE SPATIAL BIOMARKERS IN IMMUNOTHERAPY- AND TARGETED THERAPY-TREATED MELANOMA PATIENTS

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Background There is an urgent need for more robust methods to differentiate immunotherapy responders from non-responders. In this study, by utilizing a novel multiplex imaging (MI)-based panel and analysis pipeline, we found that the spatial distribution and function of immune cells correlate with response in a cohort of immunotherapy-treated melanoma patients.

Methods We designed a 28-plex sequential immunofluorescence panel (seqIF™) on the COMET™ platform¹ to target key biomarkers associated with tumor microenvironment composition (TME), immune cell infiltration, and immune checkpoint pathways. Utilizing Nucleai's deep-learning-based analysis pipeline,² we profiled 42 pre-treatment biopsies from the SECOMBIT trial³⁻⁵ (NCT02631447). We identified 15 cell types, including 10 different immune cell populations, and 10 cell state markers. Following cell typing, cellular neighborhoods were designated as previously described⁶ and were assigned to tumor or stromal areas. By analyzing 1,941 spatial features within each treatment arm, we identified the top 3 spatial features associated with progression-free survival (PFS), overall survival (OS), and prolonged clinical benefit. To perform the survival analysis, each spatial feature was binarized by proximity to the 0.25 and 0.75 quantiles, and the survival of the resulting groups was compared using the logrank test.

Results In Arm A (MAPKi inhibitor, MAPKi, until progression followed by immune checkpoint blockade, ICB), several spatial interactions were significantly associated with patient outcomes. The spatial features associated with good MAPKi outcome showed activation of the anti-tumor immune response, including enhanced antigen presentation and PD-L1 positivity within CD8 T-cells and antigen-presenting cells, and ICOS positivity within CD4 T-cells. The high content of stromal cells was associated with a worse outcome.

In Arm B (ICB until progression followed by MAPKi), PD-1+ CD8 T-cells presence in the tumor invasive margin and their interaction with PD-L1+ CD4 T-cells were associated with improved PFS and OS. High proportions of TCF1+ CD4 Tregs interactions with macrophages and T-cell interactions with endothelial cells were associated with worse outcome.

In Arm C (8 weeks of MAPKi, followed by ICB until progression, followed by MAPKi), PD-L1+ and VISTA+ antigen-presenting cells' interaction with CD4+ and CD8+ T-cells within the tumor invasive margin was associated with better

outcome. In contrast, CD8+ T-cells and CD4+ T-cells interaction with CD163+ macrophages in the outer TME was associated with a worse outcome.

Conclusions Our data demonstrates that area-specific immune niches contribute to the success or failure of immunotherapy response, highlighting the importance of spatial biology in predicting outcomes.

Acknowledgements The authors thank the patients and families who made this trial possible. Additionally, the authors acknowledge the clinical study teams and CRO who participated in the trial and in particular Paola Schiavo e Mariarita Arenella from CRT (Clinical Research Technology - Salerno). We thank Bristol-Myers Squibb (Princeton, NJ) and Array Bio-pharma Inc/Pfizer (Boulder, CO) for support. Moreover, the authors thank the participating investigators who did not enroll any patients and thus are not included as authors on the paper, Koelblinger P, Hafner C, Hoeller C (Austria), Weide B (Germany), Larkin J, Lorigan P (UK).

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Ethics Approval This study was designed in 2015 as a phase II, open-label randomized trial with no formal comparative test and a single-stage design for each arm. Patients were enrolled at 37 academic medical centers in 9 countries. The trial protocol was approved by the appropriate ethics body at each participating institution and is available in the Supplementary Information file. An independent data monitoring committee oversaw the trial. SECOMBIT is registered at ClinicalTrials.gov (NCT02631447). The study design and conduct complied with all current regulations regarding the use of human study participants and was conducted in accordance with the criteria set by the Declaration of Helsinki.

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0528>

FIRST-IN-HUMAN PHASE 1/2 STUDY OF THE INTEGRIN α V β 8-BLOCKING ANTIBODY CRB-601 IN PATIENTS WITH SOLID TUMORS

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Background Increased transforming growth factor beta (TGF β) signaling in the tumor microenvironment is associated with immune cell exclusion and poor clinical outcomes. CRB-601 is a humanized monoclonal antibody that targets integrin α v β 8, preventing activation of the latent TGF β complex. In preclinical studies, CRB-601 exhibited dose-dependent antitumor activity, decreased TGF β -1 signaling in the tumor microenvironment and enhanced the efficacy of programmed cell death protein-1 (PD-1) inhibitors.¹⁻³ Here, we report initial results from the first-in-human phase 1/2 study of CRB-601 (NCT06603844).

Methods Adults with selected advanced or metastatic solid tumors and progressive disease after ≥ 1 line of therapy were enrolled in the dose-escalation stage of the trial. Patients received CRB-601 monotherapy at doses of 3, 10 or 30 mg/kg once every 2 weeks during 28-day cycles. A Bayesian Optimal Interval design was used to determine the maximum tolerated dose and/or the pharmacologically active dose range. Safety, antitumor activity, baseline α v β 8 target levels and pharmacokinetics (PK) were also assessed.

Results As of May 12, 2025, 25 patients had enrolled in the dose-escalation phase of the study and received CRB-601 (3 mg/kg, n=8; 10 mg/kg, n=10; 30 mg/kg, n=7), of whom 13 (52%) were female and 19 (76%) were White. The mean (standard deviation) age was 62.5 (± 12.19) years, and the median (range) number of previous lines of therapy was 3 (1-9). The most common tumor types were head and neck squamous cell carcinoma (n=6), melanoma and ovarian cancer (each n=4). A total of 83 treatment-emergent adverse events (TEAEs) were reported in 21 patients. Nausea, anemia and vomiting were the most common TEAEs (all $\geq 15\%$ of patients). Grade 3 TEAEs were reported in 8 patients, with only one event, pruritus, deemed related to CRB-601. There were no grade 4 or 5 TEAEs, or dose-limiting toxicities. Antitumor activity, biomarker and PK data will be presented at the congress.

Conclusions Early phase 1/2 results suggest that CRB-601 is well tolerated with a favorable safety profile. These initial results suggest that further dose exploration with CRB-601 either alone or in combination with a PD-1 inhibitor is warranted.

Trial Registration ClinicalTrials.gov NCT06603844

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exhibits potent antitumor activity [Poster]. Presented at the 38th meeting of the Society for Immunotherapy of Cancer, Nov 1-5, 2023, San Diego, CA, USA.

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Ethics Approval Nebraska Hematology Oncology WCG 20243082 START - San Antonio Salus IRB START2023.01 Sarah Cannon - SCRI Oncology Partners Castle IRB IORG0010151 Sarah Cannon - OU Health Stephenson Cancer Center Castle IRB IORG0010151 AdventHealth Oncology Hematology WCG 20243082 Sarah Cannon - Florida Cancer Specialists/Lake Nona Castle IRB IORG0010151 University Hospital Cleveland University Hospitals IRB 20241088 Cedars-Sinai Medical Center Cedars-Sinai Office of Research Compliance and Quality Improvement 167906 UC San Diego WCG 20243082 The Clatterbridge Cancer Center NHS Foundation Trust UK London Bridge Research Ethics Committee 24/LO/0802 The University of Edinburgh - Western General Hospital - Edinburgh Cancer Research UK Centre London Bridge Research Ethics Committee 24/LO/0802

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INTRATHECAL PD-1 INHIBITOR COMBINED WITH PEMETREXED FOR LEPTOMENINGEAL METASTASES FROM SOLID TUMORS: PRELIMINARY RESULTS FROM A PHASE I/II STUDY (NCT 06462222)

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Background This study evaluated the safety and feasibility of intrathecal programmed death 1 (PD-1) inhibitor plus pemetrexed for leptomeningeal metastases (LM) from solid tumors.

Methods In phase I, a 3+3 dose de-escalation design determined dose-limiting toxicity (DLT) and the recommended dose for phase II. Patients received intrathecal pemetrexed (15 mg flat dose) twice weekly for 2 weeks (induction), then weekly for 4 weeks (consolidation), followed by monthly maintenance. Intrathecal PD-1 inhibitor (40 mg initial dose) was given every 2 weeks for 6 weeks (induction/consolidation), then monthly during maintenance. Selected patients received involved-field radiotherapy. Primary endpoints were recommended dose and safety based on adverse events (AEs). Clinical response rate (CRR), disease control rate (DCR) and overall survival (OS) were evaluated.

Results From 6/2024 to 3/2025, 31 patients were enrolled (26 non-small cell lung cancer, 4 breast cancer, and 1 small cell lung cancer). At the initial dose level of phase I, 7 patients were enrolled. No DLT was observed. The recommended dose was intrathecal 40 mg PD-1 inhibitor plus 15 mg pemetrexed. All patients completed induction therapy, 27 completed consolidation therapy, and 28 received maintenance therapy. 14 patients received radiotherapy. No treatment-related deaths occurred. Grade ≥ 3 AEs occurred in 52% (16/31), primarily hematologic toxicity (n=14) and elevated hepatic aminotransferases (n=3). Grade 1–2 immune-related AEs occurred in 32% (10/31), including rash (n=5), fever (n=3), hypothyroidism (n=1), and pneumonia (n=1). CRR was 61% (19/31) per Response Assessment in Neuro-Oncology proposal criteria, including 8 neurological dysfunction improvements, 10 CSF cytological responses, and 11 neuroimaging improvements. 74% (23/31) of patients showed LM-related neurological symptoms improvement. DCR was 87% (27/31). As of 20 June 2025, 6 patients died. Median follow-up time was 5.9 (2.0–12.1) months. Median OS was not reached.

Conclusions Intrathecal PD-1 inhibitor plus pemetrexed showed feasibility for LM from solid tumors with potential activity.

Trial Registration ClinicalTrials.gov Identifier: NCT 06462222

Ethics Approval This phase I/II study was conducted according to the Declaration of Helsinki, and was approved by the Institutional Review Board of the Affiliated Huizhou Hospital, Guangzhou Medical University, Huizhou, China (Approval number: 2024-KY-029-01). Informed consent was obtained from all subjects. (ClinicalTrials.gov Identifier: NCT 06462222)

Consent Written informed consent was obtained from the patient for publication of this abstract and any accompanying images. A copy of the written consent is available for review by the Editor of this journal.

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PHASE I STUDY: CD39⁺CD103⁺CD8⁺ SELECTED TIL THERAPY (AGX-148) DEMONSTRATES CLINICAL ACTIVITY IN PATIENTS WITH METASTATIC SOLID TUMORS

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Background Standard tumor-infiltrating lymphocyte (TIL) therapies use heterogeneous T cell populations that include both tumor-reactive and bystander cells. CD39⁺CD103⁺ double-positive (DP) CD8⁺ T cells are a phenotypically exhausted yet highly enriched subset of tumor-reactive lymphocytes within solid tumors. Despite their exhaustion markers, these cells retain potent cytotoxic activity when expanded ex vivo. AGX-148 is a selected TIL product designed to selectively isolate and expand tumor-reactive CD8⁺ T cells.

Methods This ongoing Phase I dose-escalation trial (NCT05902520) is evaluating AGX-148 in adults with metastatic solid tumors that have progressed after standard and experimental therapies. Patients undergo tumor harvest, lymphodepleting chemotherapy, adoptive transfer of AGX-148 and high-dose IL-2 for one week, with subsequent low-dose IL-2 for 1, 2, or 3 weeks depending on cohort. Primary endpoints are safety and determination of the optimal biological dose; secondary endpoints include clinical response, T cell persistence, and immune correlates.

Results AGX-148 manufacturing consistently yields billions of enriched tumor-reactive TIL. There have been no dose-limiting toxicities from AGX-148 encountered thus far. Toxicity has been as anticipated from lymphodepleting chemotherapy and IL-2. Clinical activity has been observed in solid tumors including melanoma, thyroid cancer, and oral head and neck squamous cell carcinoma with responses lasting over 600 days in some patients. TCR sequencing demonstrated long-term persistence of the infused product T cell clones in peripheral blood and metastatic tumor sites, confirming successful engraftment and tumor infiltration.

Conclusions AGX-148 offers a novel, biomarker selection strategy enriching for tumor-reactive CD8⁺ T cells for adoptive transfer. Preliminary data show clinical responses with tolerability and sustained DP TIL persistence. These findings support continued clinical development of AGX-148 and highlight the therapeutic potential of targeting tumor-specific T cell subsets in solid tumors. Updated safety, efficacy, and correlative immune data will be presented.

Trial Registration This trial is registered at ClinicalTrials.gov (NCT05902520).

Ethics Approval This clinical trial was approved by the Providence Institutional Review Board (Protocol #2023000082) and conducted under FDA IND#029-111. All participants provided written informed consent before study participation.

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TRANSLATIONAL INSIGHTS FROM A FIRST-IN-HUMAN STUDY OF NX-1607, AN ORAL CBL-B INHIBITOR, IN ADVANCED SOLID TUMORS

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Background Casitas B-lineage lymphoma proto-oncogene B (CBL-B) is an E3 ubiquitin ligase that functions as a cytoplasmic immune checkpoint, constraining immune cell activation. NX-1607 is a first-in-class, orally bioavailable, small-molecule inhibitor of CBL-B with the potential to enhance antitumor immunity. In preclinical models, NX-1607 effectively inhibits tumor growth by reshaping the intratumoral innate and adaptive immune response.^{1 2} NX-1607-101, a Phase 1 first-in-human study, demonstrated that NX-1607 exhibits a tolerable safety profile and is associated with radiographic and biomarker signals of tumor regression in patients, including tumor volume and prostate-specific antigen (PSA) reductions.

Methods As of 25 Feb 2025, 75 heavily pretreated patients with advanced solid tumors (including 19 with metastatic castration-resistant prostate cancer [mCRPC]) were enrolled in the Phase 1 study. NX-1607 was administered at doses ranging from 5 mg once daily to 40 mg twice daily. Translational analysis of pharmacodynamics (PD) included assessment of the proximal biomarker phospho-HS1 (pHS1) in CD8⁺ T cells, circulating chemokines and cytokines, and transcriptomic profiling of paired (baseline vs on-treatment) peripheral blood mononuclear cells (PBMCs) from pan-tumor and prostate cancer patients specifically, and its relationship with pharmacokinetics (PK).

Results NX-1607 exhibited dose-dependent PK with an increase in pHS1/CD8⁺ T cells. Post-treatment upregulation of circulating CXCL10 and CCL4 was observed, consistent with reported anti-PD-1 response profiles.^{3 4} RNA-seq on PBMCs from pan-tumor patient samples showed approximately 2 log₂ enrichment of immune modulation-related pathways, including IFN-γ signaling and antigen presentation. In the mCRPC patient subset, specific PBMC transcriptomic profiling revealed upregulation of innate immune responses and lymphocyte proliferation, including higher granulocyte gene expression, complement activation, cell proliferation, and cell cycle regulation. PSA reductions of ≥30% were observed in 6 out of 14 evaluable mCRPC patients. At higher doses of NX-1607, a greater reduction in PSA was associated with a trend toward increased activation of pHS1.

Conclusions NX-1607 demonstrates dose-dependent increases in PK with corresponding peripheral immune activation and preliminary antitumor signals observed in advanced solid tumors. PD data are consistent with the peripheral upregulation of both innate and adaptive immune activation pathways after NX-1607 treatment. These findings suggest that NX-

1607 has the potential to modulate the tumor microenvironment and elicit antitumor immune responses in heavily pretreated patients with solid tumors. Exploratory biomarker analyses are ongoing to further evaluate PD and to inform on mechanisms of action and resistance to NX-1607.

Trial Registration NCT05107674

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<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0593>

PHASE 2 TRIAL OF TIL THERAPY FOR METASTATIC UVEAL MELANOMA: EVALUATING CELLULAR POTENCY AND TUMOR TRANSCRIPTOMIC PREDICTORS OF RESPONSE

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Background Uveal melanoma (UM) is a rare eye cancer that frequently metastasizes to the liver and responds poorly to current immune and targeted therapies. We previously reported that adoptive cell therapy using autologous tumor-infiltrating lymphocytes (TIL) could achieve a 35% objective response rate (ORR) in metastatic UM (NCT01814046).¹ The infusion product potency, based upon autologous anti-tumor reactivity, was the strongest correlate of response. Here, we present interim results from a follow-up phase 2 trial designed to validate these findings and prospectively evaluate predictors of response.

Methods In this single-center phase 2 trial (NCT03467516), metastatic UM patients underwent metastatectomy to generate early TIL cultures. Cultures demonstrating autologous anti-tumor reactivity and growth potential were expanded for infusion. Patients received non-myeloablative lymphodepletion with cyclophosphamide and fludarabine, followed by infusion of TIL and high-dose interleukin-2. The primary endpoint was ORR using RECIST v1.1. Exploratory analyses included pooled comparison of TIL potency versus clinical response from both NCT03467516 and NCT01814046, along with transcriptomic analysis of source metastases using a recently defined in situ tumor biomarker (UMIS).²

Results As of June 2, 2025, 31 UM patients received TIL therapy. Most had high disease burden: 81% stage M1b/M1c, 77% with elevated LDH, and 71% had both hepatic and extrahepatic metastases. The median number of prior metastatic therapies was two, including checkpoint blockade in 61% and tebentafusp in 32% of patients. Liver metastases were the most common TIL source (52%). Infusion products contained a median of 5.24×10^{10} TIL, were predominantly CD8+ (median 61%), and demonstrated autologous anti-tumor reactivity in 62%. Among 30 evaluable patients, the initial overall ORR was 33% (10/30), with a confirmed ORR (cORR) of 20% (6/30) at the time of reporting. Median

duration of confirmed responses was 10.8 months (longest ongoing at 52 months). Pre-planned immunologic analysis found infusion of tumor-reactive TIL yielded a cORR of 33% (6/18), compared to 0% (0/11) with nonreactive products (figures 1 and 2). Further, in the pooled cohort (n=46), infusion of tumor-reactive TIL demonstrated significantly higher cORR of 41% (11/27) versus 0% (0/19) with nonreactive products (P=0.001). UMIS transcriptomic profiling of source metastases predicted early TIL reactivity (n=50; P<0.0001) and clinical response (n=48; P=0.008).

Conclusions This study confirms that TIL therapy has efficacy against highly advanced and pretreated metastatic UM. TIL potency, defined by autologous anti-tumor reactivity, was the strongest predictor of response. Integration of transcriptomic biomarkers such as UMIS to identify metastases harboring potent TIL may optimize patient selection and improve outcomes.

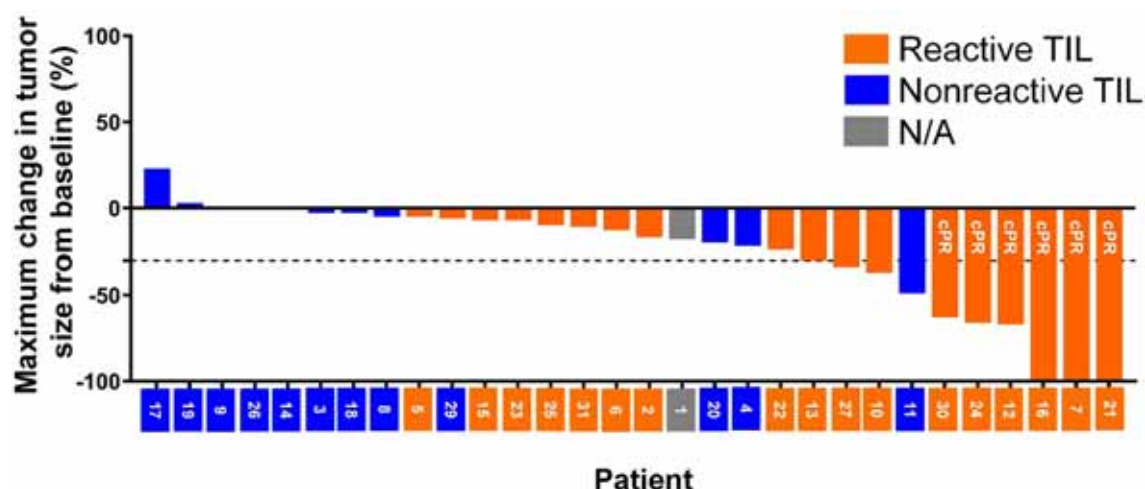
Acknowledgements Aldesleukin was provided by Clinigen Limited and Iovance Biotherapeutics.

Trial Registration NCT03467516

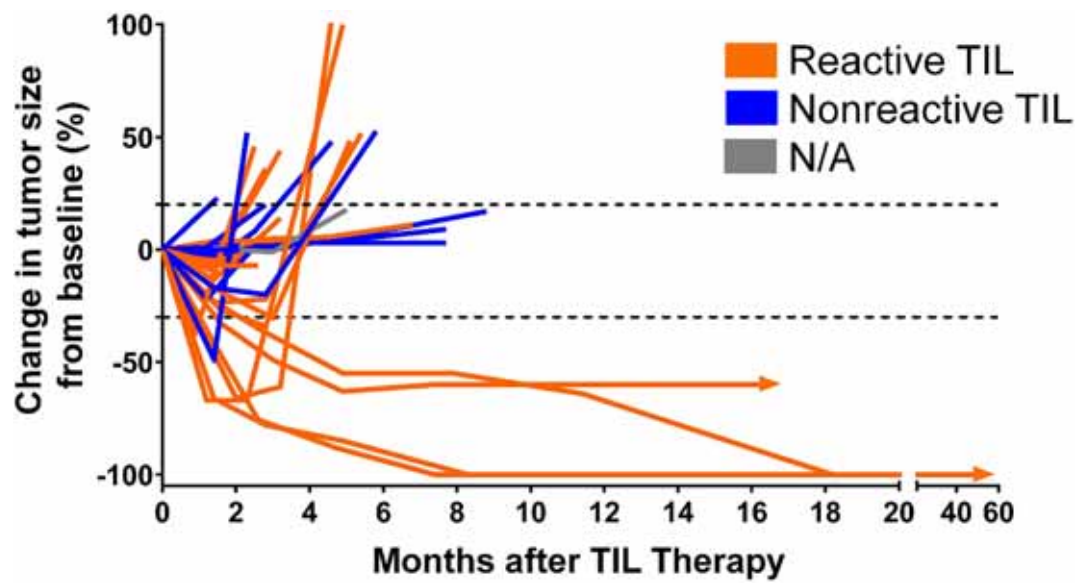
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Ethics Approval Patients gave informed consent prior to trial participation in accordance with the Declaration of Helsinki. This trial was reviewed and approved by the University of Pittsburgh Institutional Review Board (Approval CR19110103-018).



Abstract 598 Figure 1 Waterfall plot of clinical responses in patients with metastatic uveal melanoma after TIL therapy (NCT03467516)



Abstract 598 Figure 2 Spider plot of clinical responses in patients with metastatic uveal melanoma after TIL therapy (NCT03467516)

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0598>

THE MATRIX TRIAL: A MULTICENTER, RANDOMIZED, PHASE II STUDY OF ATR INHIBITION (VIA TUVUSERTIB) WITH OR WITHOUT AVELUMAB IN PATIENTS WITH ADVANCED ANTI-PD-(L)1-REFRACTORY MERKEL CELL CARCINOMA

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Background Merkel cell carcinoma (MCC) is a rare neuroendocrine skin cancer driven by UV-induced mutations or the Merkel cell polyomavirus. It is aggressive, with a high Ki-67 proliferative index. Despite an ~55% response rate to PD-1 pathway inhibitors, most patients develop primary or acquired resistance. ATR (ataxia telangiectasia and Rad3-related) kinase, a critical cell cycle checkpoint regulator, ensures genome fidelity in cancer cells experiencing high replication stress. Recent preclinical^{1,2} and clinical³⁻⁶ data suggest that ATR inhibition can enhance anti-tumor immunity and synergize with anti-PD-(L)1 therapy. Our preclinical findings suggest transcriptional induction of NF-κB-associated proinflammatory mechanisms with ATR inhibition. The potent, selective, oral ATR inhibitor tuvusertib has shown antitumor activity in phase I trials.⁶ We

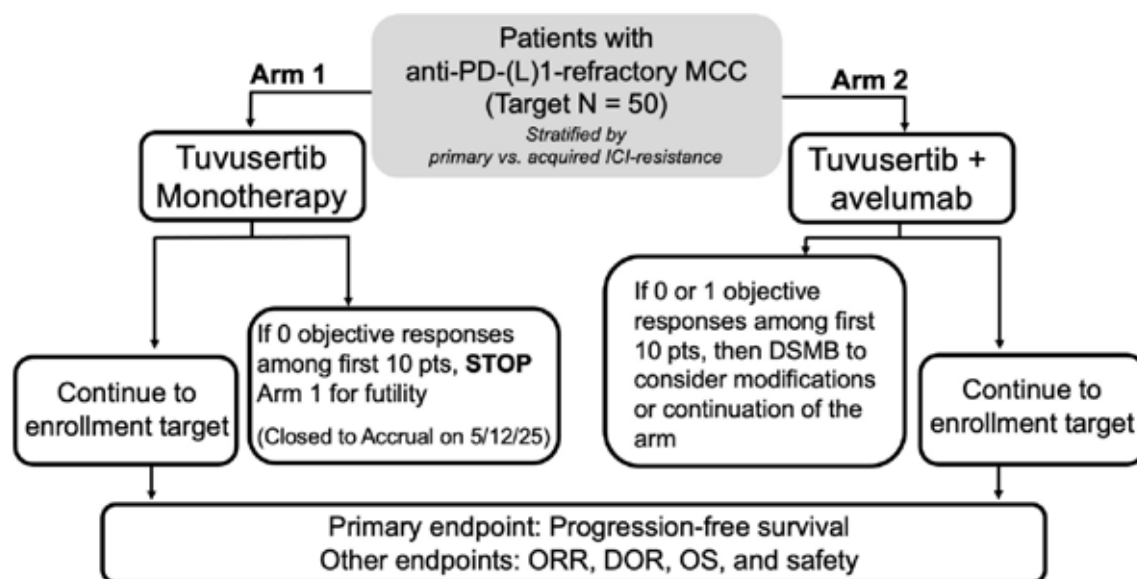
hypothesize that tuvusertib with anti-PD-(L)1 therapy, may induce tumor regression in advanced anti-PD-(L)1-refractory MCC.

Methods The multicenter, randomized, phase II MATRiX trial tests the safety and efficacy of tuvusertib ± avelumab in patients with metastatic MCC refractory to PD-(L)1 blockade (figure 1). Patients with progressive disease per Response Evaluation Criteria in Solid Tumors (RECIST v1.1) within 120 days of their last anti-PD-(L)1 therapy are eligible. Subjects in Arm 1 or 2 receive tuvusertib 180 mg QD on days 1-14 of each 21-day cycle (figure 2). Subjects in Arm 2 also receive avelumab 1600 mg IV on day 1 of each cycle. Imaging at week 9 and every 12 weeks thereafter is assessed per RECIST v1.1. The primary endpoint is progression-free survival (PFS). With a targeted enrollment of 50 patients, this trial has 83% power (one-sided level of 10%) to detect a hazard ratio of 2.0 for PFS using a stratified (primary vs. acquired resistance) log-rank test. Binary outcomes will be compared via a Mantel-Haenszel test. Tumor, blood, and stool specimens will be profiled for multi-omic signatures of immune-mediated therapeutic outcomes.

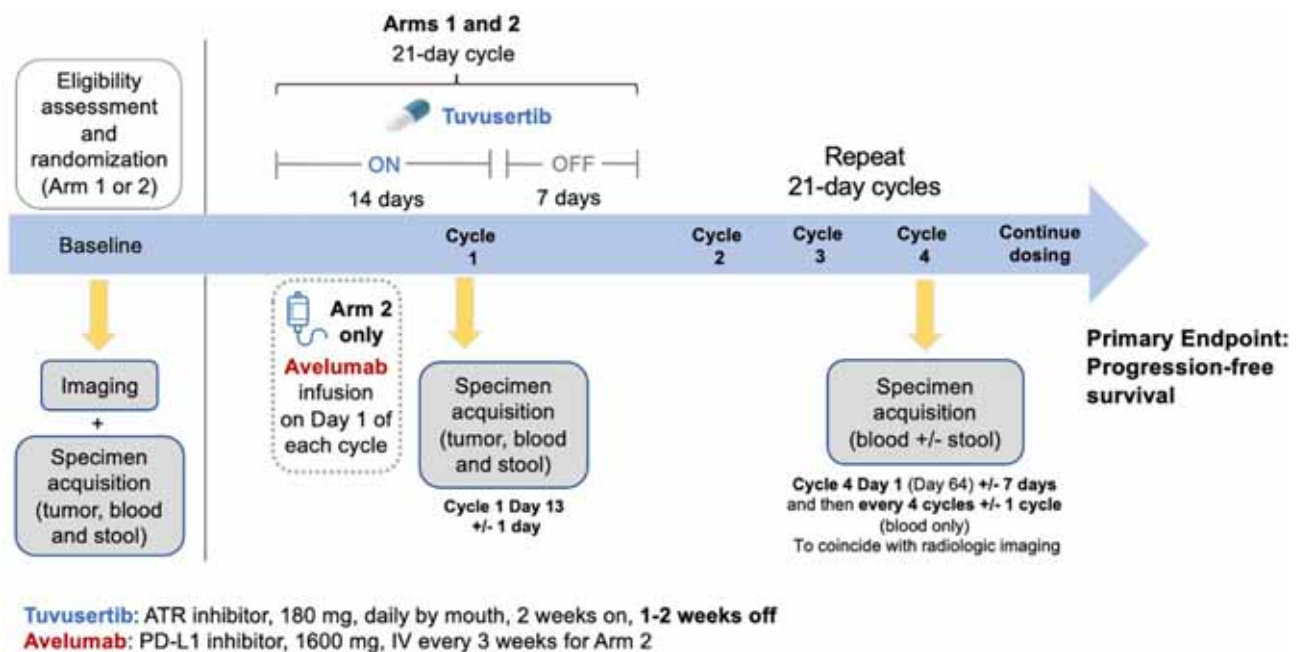
Results Between June 2024 and June 2025, 24 subjects were enrolled across 15 centers. Arm 1 (tuvusertib monotherapy) enrolled 10 patients and was closed for lack of efficacy on 5/12/2025, as per the pre-defined statistical plan; 5 crossed over to Arm 2. All additional new patients will be enrolled on Arm 2, with a target of 25 subjects.

Conclusions There is a need for additional therapeutic approaches for immunotherapy-refractory MCC, as demonstrated by robust enrollment on this trial. This orthogonal approach to solid tumor immunotherapy, relevant to more common immunogenic cancers, may guide future combination strategies to better harness anti-tumor immunity.

Acknowledgements NCI Experimental Therapeutics Clinical Trials Network, protocol writing, and regulatory teams. Patients, families, and partnering teams for their commitment to advancing our understanding of this disease. EDDOP Leadership Award-P30 Administrative Supplement, the Mark



Abstract 599 Figure 1 An overview of the multicenter, NCI-sponsored Phase II study designed to test the efficacy of tuvusertib, with or without avelumab, to address anti-PD-(L)1 resistance in advanced, refractory MCC; NCT05947500



Abstract 599 Figure 2 Timeline of treatment regimen and biospecimen acquisition for patients randomized to arm 1 and 2

Foundation for Cancer Research, the Kelsey Dickson Team Science Courage Research Team Award, and the MCC Patient Gift Fund. This study is financially supported by EMD Serono (CrossRef Funder ID: 10.13039/100004755), which provides avelumab and tuvusertib.

Trial Registration NCT05947500

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Ethics Approval The study is reviewed by NCI's Central Institutional Review Board; approval number is 10592.

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0599>

RETREATMENT WITH RP1 IN COMBINATION WITH NIVOLUMAB IN PATIENTS WITH ADVANCED ANTI-PD-1-FAILED MELANOMA

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Background Treatment options for melanoma that has progressed on anti-PD-1 therapy are limited. Vusolimogene oderparepvec (RP1) is a fusion-enhanced HSV-1-based oncolytic immunotherapy. The IGNYTE trial showed that RP1 in combination with nivolumab (nivo) demonstrated clinically meaningful, durable responses (objective response rate, 32.9%; median duration of response, 33.7 months by RECIST 1.1) in patients with anti-PD-1-failed melanoma, with mainly grade 1/2 constitutional-type side effects. The trial allowed retreatment with RP1 for patients deriving clinical benefit. Here, we present the investigator assessment of patients who were retreated with RP1 in the IGNYTE trial.

Methods Patients with melanoma and confirmed progression on anti-PD-1 ± anti-CTLA-4 therapy for ≥8 weeks as the last prior treatment were enrolled. During the initial course, RP1 was given intratumorally at 1×10⁶ PFU/mL initially, then at 1×10⁷ PFU/mL Q2W (≤7 doses) with intravenous nivo (240 mg). Nivo was then given alone (240 mg Q2W or 480 mg Q4W) for 2 years. Retreatment with RP1 could occur if patients had injectable lesions, stable performance status, and no treatment-related grade 4 adverse events (AEs) or grade 3 injection-site reactions, infections, or immune-mediated AEs lasting >14 days. Additional doses of RP1 were administered at 1×10⁷ PFU/mL Q2W either in combination with nivo or as monotherapy if nivo was stopped due to nivo-related toxicity. An investigator survey was performed to assess efficacy and safety in patients who received retreatment with RP1.

Results Of 140 enrolled patients, 25 (18%) were retreated with RP1 after the first course of therapy (range, 1–24 additional doses). Of these, 17 patients experienced clinical benefit as determined by the investigator. Reasons for further treatment included aiming for deepening of response or delay of disease progression. Observed anti-tumor effects included complete resolution of new lesions (n = 1 patient) and deepened response (n = 4 patients). Five patients reported at least one

grade ≥3 treatment-emergent AE, most of which were deemed not related to RP1.

Conclusions Retreatment with RP1 demonstrated investigator-assessed clinical benefit in the majority of patients to whom it was given (17/25 patients; 68%). Retreatment was well tolerated with the safety profile being consistent with that of the study overall. Additional clinical data from this collection of case reports will be presented.

Trial Registration ClinicalTrials.gov NCT03767348

Ethics Approval The study was conducted in accordance with the ethical principles originating from the Declaration of Helsinki and was approved by the institutional review board/ethics committee at each participating site. Written informed consent was obtained from all patients prior to the conduct of any study-related procedures.

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0600>

ASSESSMENT OF CRISPR PD-1 EDITED TUMOR-INFILTRATING LYMPHOCYTES IN INFUSION PRODUCTS FOR ADOPTIVE CELL THERAPY IN METASTATIC MELANOMA

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Background Adoptive T cell therapy with expanded tumor-infiltrating lymphocytes (TILs) and PD-1 targeting immune checkpoint inhibitors has shown the potential to induce long-lasting responses in metastatic melanoma (MM), although such responses remain limited. Genetic editing of TILs, such as PD-1 knockout, offers a promising strategy to enhance antitumor efficacy by rendering these cells resistant to tumor-driven immunosuppression. We previously established that non-viral delivery of CRISPR-Cas9 can efficiently knockout PD-1 in human TILs.¹ Here, we present initial data from the first-in-human trial employing CRISPR-Cas9 to knockout PD-1 in TILs from patients with metastatic melanoma (NCT06783270).

Methods The CRISPR-TIL infusion products were phenotypically characterized by flow cytometry. Antitumor reactivity was assessed by measuring the TCR-mediated recognition of autologous tumor antigens through the expression of IFN- γ , TNF- α , CD137, and CD107a following co-culture. Additionally, the cytotoxic capacity of an infusion product was evaluated using a real-time target cell killing assay.

Results Four patients received CRISPR-edited TIL infusion products. The infusion products had an efficient PD-1 knockout and maintained their cellular composition, with CD3⁺ subsets predominantly comprising effector memory T cells. The infusion product from patient 02 was evaluated for antitumor reactivity and found to contain tumor-reactive TILs, as demonstrated by cytokine production and the upregulation of CD107a and CD137 when co-cultured with autologous tumor cells. Cytotoxicity assays confirmed robust killing of autologous tumor cells and sustained growth control during a 72-hour co-culture period.

Conclusions These preliminary data from the first-in-human trial of CRISPR-edited TILs demonstrate the feasibility of non-viral PD-1 knockout with high editing efficiency. Functional assays indicate that gene-edited TILs retain potent antitumor activity. Further investigation is warranted to evaluate clinical efficacy and durability of response.

Trial Registration 2023-510417-25-01 (EU Trial (CTIS) Number) NCT06783270 - clinicaltrials.gov

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Ethics Approval The Danish Medicines Agency and The Danish National Committee on Health Research Ethics approve the study. All participants gave oral and written informed consent before taking part 2023-510417-25-01 (EU Trial (CTIS) Number).

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0601>

RESTORING IMMUNE RESPONSIVENESS IN PERIPHERY AND TUMOR WITH MRNA-4359, AN MRNA-BASED THERAPY ENCODING PD-L1 AND IDO1 PEPTIDES, PLUS PEMBROLIZUMAB IN CHECKPOINT INHIBITOR RESISTANT/REFRACTORY MELANOMA

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Background mRNA-4359 is an investigational immunotherapy designed to activate antigen-specific T cells that target both PD-L1 and IDO1 expressed by tumor and immunosuppressive cells. By promoting T-cell responses and reshaping the tumor microenvironment (TME), mRNA-4359 may enhance the efficacy of immune checkpoint blockade in resistant settings. Here, we present peripheral and tumor translational data from checkpoint inhibitor resistant/refractory (CPI-R/R) melanoma patients (pts) treated with mRNA-4359 plus pembrolizumab.

Methods In an ongoing trial (NCT05533697), as of 28Feb2025, 29 patients with CPI-R/R melanoma received pembrolizumab (400 mg IV Q6W) plus mRNA-4359 (400 µg [n=14] or 1000 µg [n=15] IM, Q3W, up to 9 doses. Peripheral and tumor samples collected pre- and on-treatment were analyzed for antigen-specific T cell responses, T cell clonality, ctDNA, and features of an immune-inflamed TME.

Results Twenty-five pts were response-evaluable with objective response rate (ORR) of 24% (n = 6, 1 complete responder, CR; 5 partial responders, PR) and disease control rate (DCR) of 60% (n=15: 6 ORRs + 9 stable disease, SD). Peripheral PD-L1- or IDO1-specific T cell responses were detected post-treatment after *in vitro* expansion in majority of pts with evaluable samples (11/16, 69%). Treatment led to increases of novel expanded T cell receptor (TCR) clones over time across both dose groups (n=23). The median peak number of novel expanded TCR clones was higher in responders, with clone counts of 82 in CR (n=1), 67 in PR (n=5), 24 in SD (n=8), and 22 in progressive disease (PD, n=9), assessed up to 269 days post-treatment. A few peripherally expanded TCR clones were also detected in responder tumors 3 weeks after the first dose, and most responders (4/5) show >95% reduction in ctDNA tumor methylation score at week 6.

Among 23 pts with baseline biopsy, dual gene expression of PD-L1 and IDO1 was higher in responders, which correlated with an immune inflamed phenotype. On-treatment tumor samples showed increases in immune-inflamed gene signatures related to T cell inflammation and antigen presentation, which were ≥2-fold higher in PR (n=4) compared to pts with SD or PD (n=11). Responders (n=4) exhibited greater increases in T cell infiltration compared to non-responders (n=11), with 6.2-fold and 1.9-fold higher levels for CD4 and CD8 T cells, respectively.

Conclusions mRNA-4359 plus pembrolizumab may restore immune responsiveness in CPI-R/R melanoma by priming antigen-specific T cells, expanding de novo TCR clones, and creating a more inflamed TME with increased T cell infiltration, allowing for restoration of checkpoint inhibitor response.

Acknowledgements This study was funded by Moderna, Inc.

Ethics Approval The protocol and all amendments were approved by the appropriate institutional review board or independent ethics committee at each participating study site. The study was conducted in accordance with the principles of the Declaration of Helsinki and the International Conference on Harmonization Good Clinical Practice guidelines.

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PD-1-DIRECTED INTRATUMORAL IMMUNOTHERAPY FOR CUTANEOUS CARCINOMAS: A CLINICAL STUDY OF INTASYL PH-762

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Background Immune checkpoint-targeted antibodies directed at PD-1 or PD-L1 block co-inhibitory receptors expressed by anti-tumor T cells, breaking immune tolerance against tumor cells and generating cancer immunity. Immunotherapy by intratumoral (IT) injection aims to use the tumor as a ‘self-vaccine’, stimulating local immune activation and priming an anti-tumor immune response. This may generate abscopal tumor responses, activating untreated tumors via circulating activated anti-tumor immune cells. Local delivery of immunotherapy minimizes systemic exposure and off-target toxicities and may decrease tumor size and improve surgical morbidity.

The INTASYL™ compound PH-762 is designed to precisely silence PD-1 mRNA. INTASYL is a patented, self-delivering RNAi technology designed to impart specific properties to small interfering RNAs. INTASYL's unique structural and chemical modifications ensure optimized cell and tissue uptake of PH-762 with IT administration. The efficient uptake of PH-762 by human T cells, silencing of PD-1 mRNA and subsequent protein reduction has been demonstrated preclinically. Murine-targeted PH-762 (mPH-762) injections can silence PD-1 mRNA in T cells within the murine tumor and increase the secretion of IFN- γ .

Methods This open-label Phase 1 clinical study (NCT 06014086) is designed to evaluate the safety and tolerability of neoadjuvant use of IT PH-762 in cutaneous squamous cell carcinoma, melanoma, or Merkel cell carcinoma, to determine the pharmacokinetic profile of PH-762 after IT injection, to observe pathologic and immunologic tumor responses, and to determine the recommended dose for development. Eligible tumors are at least 1.0 cm and no more than 3.0 cm in longest dimension. Escalating dose concentrations of PH-762 (from 1.14 mg/mL through 22.00 mg/mL) are tested serially in cohorts of 3 patients each. Patients receive IT PH-762 once weekly, 4 times over a 3-week period prior to surgical excision, which occurs 5 weeks after the initial injection. Tumor changes are evaluated per iRECIST criteria and pathological response.

Trial Registration Clinicaltrials.gov: NCT 06014086

Ethics Approval This study was approved by Advarra Institutional Review Board (00023875).

<http://dx.doi.org/10.1136/jitc-2025-SITC2025.0610>

RP1 PLUS NIVOLUMAB IN PATIENTS WITH AND WITHOUT PRIOR BRAF-DIRECTED THERAPY: A SUBGROUP ANALYSIS OF PATIENTS WITH ANTI-PD-1-FAILED BRAF-MUTANT MELANOMA FROM THE IGNYTE CLINICAL TRIAL

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Background Approximately 50% of melanomas have *BRAF* mutations, which drive the proliferation and survival of tumor cells via constitutive activation of the MAPK pathway. *BRAF* inhibitors (BRAFi) ± MEK inhibitors (MEKi) may be used to treat advanced *BRAF* V600-mutant melanoma as first-line therapy or after progression on anti-PD-1 therapy. However, prior treatment with *BRAF*-directed therapy can diminish the efficacy of subsequent treatment with anti-PD-1 therapy. Vusolimogene oderparepvec (RP1; an HSV-1-based oncolytic immunotherapy) combined with nivolumab demonstrated clinically meaningful, durable antitumor activity (objective response rate [ORR], 32.9% by RECIST 1.1; median duration of response [DOR], 33.7 months) in patients with melanoma and confirmed progression on prior anti-PD-1 therapy. Here we

report the impact of prior BRAFi/MEKi therapy on the efficacy of RP1 plus nivolumab.

Methods Patients with melanoma and confirmed progression on anti-PD-1 ± anti-CTLA-4 therapy for ≥8 weeks as the last prior treatment were enrolled. Prior treatment with BRAFi ± MEKi therapy was allowed for *BRAF* V600 mutation-positive melanoma. RP1 was given intratumorally at 1×10⁶ PFU/mL initially, then at 1×10⁷ PFU/mL Q2W (≤7 doses) with intravenous nivolumab (240 mg), followed by nivolumab alone (240 mg Q2W or 480 mg Q4W) for up to 2 years. This post hoc analysis evaluated ORR, DOR, and progression-free survival (PFS) by RECIST 1.1 per independent review, and overall survival (OS) in patients with *BRAF*-mutant melanoma, stratified by prior BRAFi treatment (data cutoff: March 8, 2024).

Results Of 140 enrolled patients, 53 had *BRAF*-mutant melanoma, including 16 (30.2%) who had received BRAFi/MEKi prior to anti-PD-1 therapy (BRAFi-exposed) and 37 (69.8%) who were BRAFi-naïve. The confirmed ORR was 12.5% in the BRAFi-exposed group vs 43.2% in the BRAFi-naïve group (table 1). Median DOR was 3.9 months in the BRAFi-exposed group vs 33.7 months in the BRAFi-naïve group. Median PFS was 1.9 months in the BRAFi-exposed group vs 5.6 months in the BRAFi-naïve group; the 12-month PFS rate was 13.3% vs 40.4%, respectively. The 12-month OS was 71.4% in BRAFi-exposed patients and 85.9% in BRAFi-naïve patients. Fifteen of the 37 BRAFi-naïve patients were subsequently treated with BRAFi ± MEKi for a median of 13.9 months with an ORR of 66.7%.

Conclusions In patients with advanced *BRAF*-mutant melanoma that progressed on prior anti-PD-1 therapy, RP1 plus nivolumab appeared to be more effective in BRAFi-naïve patients. This finding is consistent with literature showing that *BRAF*-directed therapy leads to cross-resistance to subsequent immunotherapy.

Trial Registration ClinicalTrials.gov NCT03767348

Ethics Approval The study was conducted in accordance with the ethical principles originating from the Declaration of Helsinki and was approved by the institutional review board/ethics committee at each participating site. Written informed consent was obtained from all patients prior to the conduct of any study-related procedures.

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Abstract 611 Table 1 Efficacy by prior BRAFi/MEKi treatment

	BRAFi-exposed (n = 16)	BRAFi-naïve (n = 37)
ORR, % (95% CI)	12.5 (1.6, 38.3)	43.2 (27.1, 60.5)
CR, %	6.3	18.9
PR, %	6.3	24.3
SD, %	31.3	32.4
PD, %	50.0	18.9
DOR, median (95% CI), months	3.9 (NR, NR)	33.7 (9.3, NR)
PFS, median (95% CI), months	1.9 (1.8, 3.6)	5.6 (3.5, 16.2)
OS, % (95% CI)		
1 year	71.4 (40.6, 88.2)	85.9 (69.3, 93.9)
2 years	52.1 (21.8, 75.6)	76.5 (54.6, 88.8)

CR, complete response; NR, not reached; PD, progressive disease; PR, partial response; SD, stable disease.

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A CLINICAL TRIAL TO EVALUATE THE SAFETY AND PRELIMINARY EFFICACY RESULTS OF CAR-MACROPHAGE IN METASTATIC HER2-LOW EXPRESSING AND HER2-POSITIVE SOLID TUMORS

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Background Macrophages possess robust tumor-infiltrating capabilities. Engineered macrophages (chimeric antigen receptor macrophages, CAR-M) with enhanced targeting specificity and the ability to evade immunosuppressive microenvironments represent a groundbreaking advancement in solid tumor therapy. The mentioned CAR-M is a third generation of CAR-M, incorporating a tandem design of cytokine GM-CSF, which is essential for macrophage autonomous proliferation and activation along with the CAR construct. This innovative approach aims to overcome the current therapeutic limitations in solid tumor management.

Methods As of June 2025, a total of 7 patients (median age 56) were enrolled, including 3 HER2-positive and 4 HER2-low cases. These patients received CAR-M infusion via intraperitoneal in a dose-escalation manner at one-week intervals. The primary endpoints included evaluating tumor progression status and clinical improvement, as well as calculating ORR, CR, and PR. The secondary endpoint was PFS.

Results CAR-M showed favorable safety and tolerability profiles. No DLTs or ICANS were observed during infusion. The incidence of $\geq$ grade 2 TEAEs was 0. Among the 7 patients, only mild CRS occurred, primarily manifesting as fever and chills, which resolved spontaneously within 48 hours. HER2-positive patients exhibited reduced tumor biomarkers and radiologically confirmed tumor shrinkage. The DCR was 100%, with a 6-month PFS rate of 100%. Notably, 5/7 patients experienced weight gain. HER2-low patients indicated tumor stabilization or regression, achieving a DCR of 50% and a median PFS of 4.6 months. Transient T-cell exhaustion was observed in one case, strikingly, however, immune recovery occurred within one month.

Conclusions CAR-M demonstrated a manageable safety profile along with encouraging response rates. Further CAR-M investigations targeting additional antigens are currently being planned.

Trial Registration ChiCTR2400082776 and ChiCTR2400080078

Ethics Approval Approval Letter of Clinical Research Ethics Committee of The Affiliated Hospital of Xuzhou Medical University, Xuzhou, China. ID: XYFY2023-KL107-01 and XYFY2024-KL077-01

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PHASE 1 STUDY OF INTRATUMORAL VAX014 WITH DOSE EXPANSION IN COMBINATION WITH PEMBROLIZUMAB OR NIVOLUMAB IN PATIENTS WITH ADVANCED SOLID TUMORS

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Background VAX014 is a novel, non-viral oncolytic immunotherapy based on recombinant bacterial minicells (rBMCs).¹ Unlike oncolytic virotherapies, VAX014 does not require BSL-2 containment and has optimal antitumor activity in tumors expressing the antiviral pattern recognition receptors, STING and/or RIG-I.² VAX014 is being evaluated for intratumoral treatment of advanced solid tumors in a two-part multi-center Phase 1 study (NCT 05901285). The dose escalation component of the study is complete and the dose expansion portion giving VAX014 in combination with investigator's choice of PD-1 inhibitor is being initiated.

Methods Eligible patients (pts) have a performance status of 0-2 (ECOG), measurable disease, and at least 1 injectable advanced solid tumor (1-10 cm in longest diameter) accessible by either physical exam (dose escalation) or ultrasound or other image-guided injection of visceral tumor(s) (dose expansion). The dose escalation segment employed a 3+3 design with 5 planned dose cohorts. Dose escalation with weekly intratumoral administration of VAX014 will proceed until a maximum tolerated dose or maximum practical dose is reached.

Dose expansion will evaluate VAX014 in combination with investigator's choice of pembrolizumab (200mg/Q3weeks) or nivolumab (240mg/Q2weeks) in patients with solid tumors that have progressed on prior PD-1 therapy.

The primary study objectives are to establish safety and tolerability. Secondary study objectives will evaluate pharmacokinetics, anti-drug antibody formation, changes in intratumoral and peripheral immune status, and response(s) in injected and non-injected lesions (per itRECIST) and overall response (per RECIST 1.1).

Results As of 26-June-2025, 15 pts with advanced solid tumors have been treated with VAX014 monotherapy over 5 dose cohorts ranging from 4.0×10^6 to 4.0×10^8 VAX014 rBMCs (10 M/8 F, median age 61 (range, 35-81), median prior therapies 5 (range, 2-16), with 12 of 15 (80%) having prior PD-(L)1 inhibitor. No DLTs have been reported, and VAX014-related toxicity has been limited to Grade 1-2 adverse events.

Thirteen of 15 (86.6%) pts had stable disease (SD) or better in injected tumors at any timepoint with 5/15 demonstrating $\geq 30\%$ reduction from baseline (-60%, -53%, -48%, -38% and -32%). Three of 15 (20%) pts had SD or better at any time in noninjected tumors with 2 demonstrating $\geq 30\%$ reduction (1 with 48% reduction, 1 completely resolved). The best overall response was SD (RECIST 1.1).

Conclusions VAX014 monotherapy is well tolerated with evidence of activity in both injected and noninjected tumors across dose levels and will now be evaluated in combination

with investigator's choice of pembrolizumab or nivolumab during dose expansion.

Trial Registration NCT 05901285

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Ethics Approval This Phase 1 study obtained ethics approval via a central Institutional Review Board (WIRB; IRB No. 20232966) and/or the Institutional Review Board of each respective institution (Dana-Farber Cancer Institute, IRB No. 23-304; Cleveland Clinic, IRB No. 24-280; Dartmouth Hitchcock, IRB No. STUDY02002393). All participants in the study gave informed consent prior to enrollment.

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INITIAL SAFETY AND EFFICACY RESULTS FROM A FIRST IN HUMAN, PHASE 1/2 STUDY OF ASKG915, AN ANTI-PD-1/PRO-IL-15 BIFUNCTIONAL FUSION PROTEIN, FOR PATIENTS WITH ADVANCED SOLID TUMORS

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Background ASKG915 is a bifunctional fusion protein of PD-1 antibody fused with an IL-15 prodrug. Preclinical studies demonstrated its efficacy in neoplastic models. Here, we present initial safety and efficacy results from a first-in-human, dose escalation and dose expansion of ASKG915 monotherapy in patients (pts) with advanced solid tumors.

Methods This open-label, multicenter, Phase 1/2 study in adult pts with advanced solid tumors (NCT05867420) comprised two parts: dose escalation and dose expansion. Pts with advanced solid tumors that were resistant/refractory to current standard treatment, and with at least 1 measurable lesion per RECIST 1.1, were eligible. The primary objective was safety. Secondary objectives included efficacy, pharmacokinetics, pharmacodynamics, and immunogenicity.

Results As of June 09, 2025, 46 pts were enrolled at dose-escalation phase (n=16) and dose expansion phase (n=30). Primary tumor types included NSCLC (n=17, 37%), colorectal cancer (n=13, 28%), ovarian cancer (n=8, 17%), cervical cancer (n=4, 9%) and others (n=4, 9%). Most pts (39/46, 85%) had received more than two lines of prior treatment, and 30 pts (65%) had undergone previous immunotherapy.

No dose-limiting toxicity was observed and the maximum tolerated dose has not been reached. Grade 3 and above ASKG915-related treatment emergency adverse event (TEAE) occurred in five pts (31.3%). Serious TEAEs were reported in four pts (25.0%). Based on overall safety data, two doses were selected for further exploring at dose expansion phase. During the dose expansion phase, anemia (9/30, 30%), rash (7/30, 23.3%) and pyrexia (6/30, 20%) were the most frequently reported TEAEs; grade 3 ASKG915-related TEAEs only occurred in three pts (10%), including abnormal liver function, rash and blood pressure elevated. All toxicities were manageable/reversible.

The ORR and DCR were 8.3% and 47.2% across all dose levels of ASKG915 monotherapy. Three partial responses were observed (2 MSS CRC and 1 NSCLC). In 3rd line and above MSS CRC subgroup at the mid dose level or above (n=10), the ORR were 20% in all pts regardless of liver metastasis (LM) status and 33.3% in pts (n=6) without LM. CD8+ T and NK cells in pts receiving mid dose at or above showed an increasing trend towards proliferation and expansion, alongside elevated levels of INF- γ . The total plasma drug exposure in pts increased dose-dependently within the range of study doses after the initial administration.

Conclusions ASKG915 monotherapy showed a tolerable safety profile and promising antitumor activity in heavily pretreated advanced solid tumors. These results support further evaluation of ASKG915.

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Trial Registration Clinical trial information: NCT05867420

Ethics Approval The protocol and all amendments were approved by the independent ethics committee at each participating study site. The study was conducted in accordance with the principles of the Declaration of Helsinki and the International Conference on Harmonization Good Clinical Practice guidelines.

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IMNESKIBART, A HUMAN MONOCLONAL ANTIBODY (MAB) THAT BINDS IL-2 AND PREVENTS CD25 BINDING, + LOW-DOSE SUBCUTANEOUS IL-2: PHASE 2 UPDATE ON CPI-REFRACTORY MELANOMA AND NON-SMALL CELL LUNG CANCER (NSCLC)

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Background Imneskibart (AU-007) is a mAb binding to interleukin-2 (IL-2) at an epitope that prevents binding to CD25 in the trimeric IL-2 receptor, while allowing engagement with the dimeric receptor, CD122/CD132. Imneskibart bound IL-2 expands T effector (Teff) and NK cell populations (dimeric IL-2R) without expanding regulatory T cells (Tregs; trimeric IL-2R). Avelumab is an anti-PD-L1 mAb containing an active Fc region. Avelumab can block the PD-L1/PD-1 checkpoint, and initiate antibody-dependent cell-mediated cytotoxicity mediated by NK cells, bringing innate and adaptive immunity against tumors.

Methods Four ongoing Phase 2 (Ph2) expansion cohorts evaluate 8-week cycles of the recommended Ph2 dose (RP2D) of 9 mg/kg imneskibart IV Q2W + one subcutaneous (SC) 135K IU/kg aldesleukin dose on Day 1. Ph2 Cohort 1 evaluates the RP2D and Cohort 4 evaluates the RP2D-1/RP2D + nivolumab (data immature, not reported), both in melanoma; Cohort 2 evaluates the RP2D and Cohort 3 evaluates the RP2D-1/RP2D + avelumab in PD-L1-positive ($\geq 1\%$) NSCLC that progressed on anti-PD-1/L1 $\pm$ chemotherapy.

Results Twelve melanoma patients enrolled in Cohort 1 and two in Ph1 (received below the RP2D). Two melanoma patients refractory to anti-CTLA-4 and anti-PD-1 had tumor reductions of 48% (14 months on treatment) and 100% (continues treatment at 16 months). A patient refractory to anti-PD-1 + anti-LAG-3 had a 58% tumor reduction (continues treatment at 13 months). An acral melanoma patient progressed on anti-CTLA-4 and anti-PD-1 had 12 months stable disease (SD).

In NSCLC, of two Cohort 2 patients, one is ongoing with SD at five months. In Cohort 3, six NSCLC patients received avelumab plus the RP2D-1 (four patients; one with 45% target lesion reduction refractory to prior anti-PD-1, two with SD, one pending evaluation) or the RP2D (two patients; one with progressive disease, one pending evaluation).

Drug related Grade 3/4 events occurred in 15% of patients receiving imneskibart plus aldesleukin, and the most common Grade 1/2 events occurred in $\leq 20\%$ of patients.

Conclusions Imneskibart + low-dose SC aldesleukin possess a unique mechanism of action by reducing Tregs while increasing the Teff/Treg ratio, resulting in clinically meaningful tumor regressions and a low incidence of Grade 3/4 adverse events. Melanoma patients with primary resistance to prior doublet

immune checkpoint inhibitors had durable, deep tumor reductions. Early signals of tumor regression were observed in the newly opened avelumab combination cohort of NSCLC patients progressing on prior anti-PD-1/L1 $\pm$ chemotherapy. Updated data, including additional patients, will be presented at the conference.

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Trial Registration NCT05267626

Ethics Approval This study was approved by Monash Health (Australia) HREC/80672/MonH-2021-295282(v2) and Advarra IRB (US) CR00561013. All patients provided informed consent prior to enrollment.

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COMBINING THE BISPECIFIC ANTIBODIES BNT327 (PD-L1xVEGF-A) AND BNT314 (EPCAMx4-1BB) ENHANCES T-CELL MEDIATED CYTOTOXICITY AND ANTITUMOR ACTIVITY IN PRECLINICAL STUDIES

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Background BNT327 (PM8002) and BNT314 (GEN1059/Duo-Body®-EpCAMx4-1BB) are investigational bispecific antibody (bsAb) immunomodulators. BNT327 combines blockade of the PD-1/PD-L1 checkpoint with VEGF-A neutralization, thereby reducing immune suppression and promoting vascular normalization, immune-cell activation and tumor infiltration. BNT314 is designed to boost antitumor immune responses through EpCAM-dependent activation of the co-stimulatory receptor 4-1BB expressed on activated T cells. Previous studies demonstrated that combining BNT314 with PD-1/PD-L1 blockade further potentiates its effects on T-cell function *in vitro* and antitumor activity *in vivo*. Here we investigated the potential of combining BNT327 with BNT314 to promote antitumor immunity in preclinical studies.

Methods Co-cultures of CD8⁺ T cells purified from human peripheral blood mononuclear cells with EpCAM-transduced MDA-MB-231 tumor cells were used to study T-cell cytotoxic activity *in vitro*. *In vivo*, Fc-silenced bsAbs targeting mouse PD-L1 and mouse VEGF-A (mPD-L1xmVEGF-A) or human EpCAM and mouse 4-1BB (hEpCAMxm4-1BB) were tested in human EpCAM-transgenic mice bearing human EpCAM-expressing MC38 (MC38-hEpCAM) colorectal or B16F10 (B16F10-hEpCAM) melanoma tumors. Tumor growth and progression-free survival (PFS, defined as time until tumor volume reaches ≥500 mm³) were determined.

Results Combining BNT327 with BNT314 enhanced expression of granzyme B and CD107a by CD8⁺ T cells, and significantly increased T-cell mediated killing of EpCAM⁺ tumor cells expressing cognate antigen *in vitro*, compared to each single agent. In this assay, PD-L1 blockade activity of BNT327 was comparable to atezolizumab both as a single treatment and in combination with BNT314. *In vivo*, combining mPD-L1xmVEGF-A with hEpCAMxm4-1BB delayed outgrowth of subcutaneous MC38-hEpCAM tumors and prolonged mouse PFS compared to both monotherapies. Combination treatment also mediated immunomodulation that was more pronounced compared to single-agent treatments, including enhanced frequencies of activated or tumor-targeting CD8⁺ T cells, and a shift towards an effector/memory CD8⁺ T-cell phenotype in the peripheral blood. Moreover, in mice bearing subcutaneous B16F10-hEpCAM tumors, the combination of mPD-L1xmVEGF-A and hEpCAMxm4-1BB enhanced antitumor activity and prolonged PFS beyond the single-agent effect of mPD-L1xmVEGF-A, whereas hEpCAMxm4-1BB alone had no effect in this model.

Conclusions Combining a PD-L1xVEGF-A bsAb with an EpCAMx4-1BB bsAb potentiates single-agent effects on T-cell cytotoxic activity *in vitro* and enhances antitumor activity *in vivo* in a colorectal cancer model, and in a poorly immunogenic melanoma model that is unresponsive to hEpCAMxm4-1BB bsAb alone. Building on these data, a clinical trial is being initiated to investigate the safety and preliminary efficacy of BNT327 in combination with BNT314 and chemotherapy in patients with metastatic microsatellite-stable/proficient mismatch repair (MSS/pMMR) colorectal cancer.

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